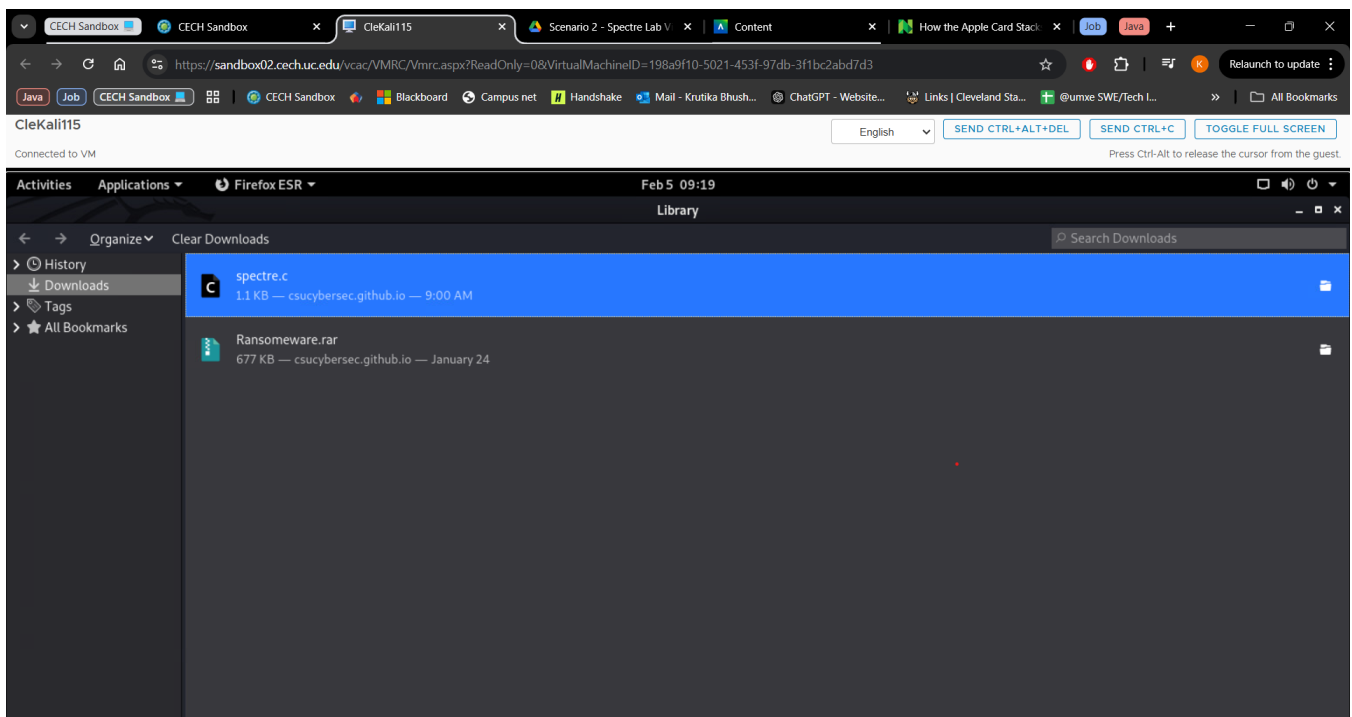


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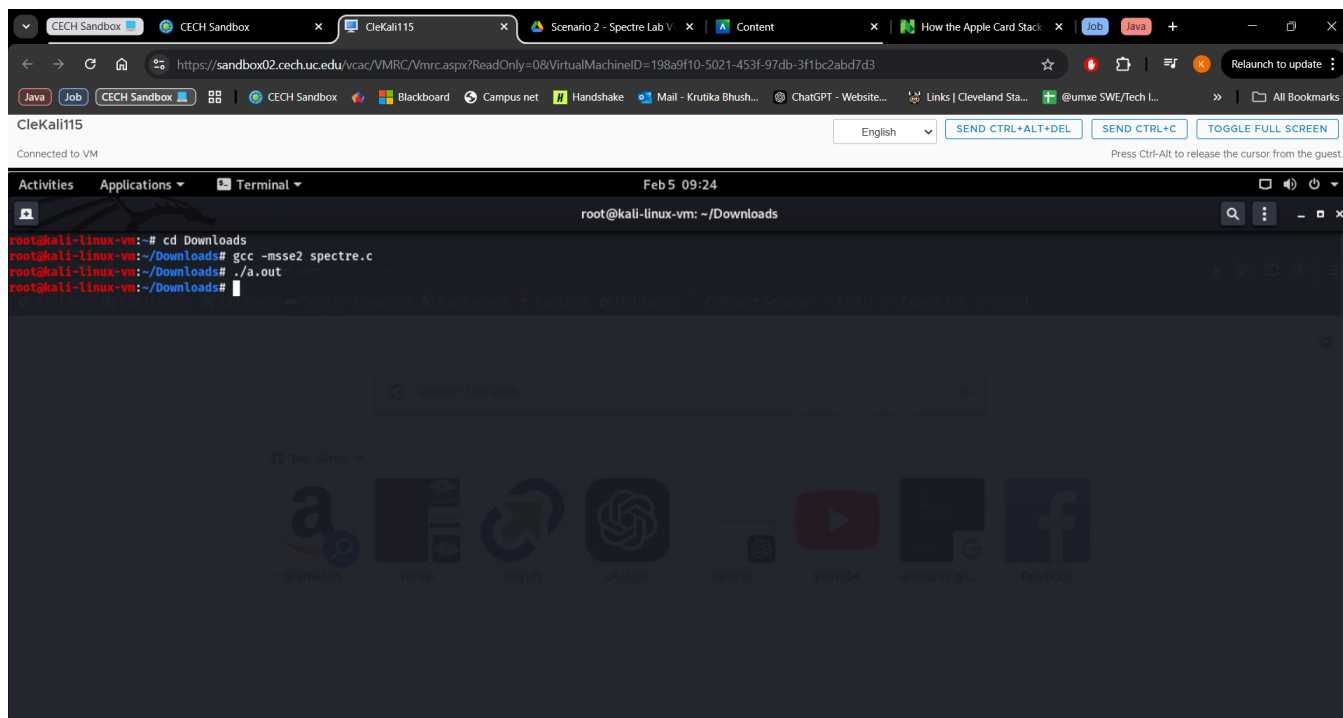
Assignment 2

Exercise 1: Scenario 2 - Spectre Attack Scenario Lab

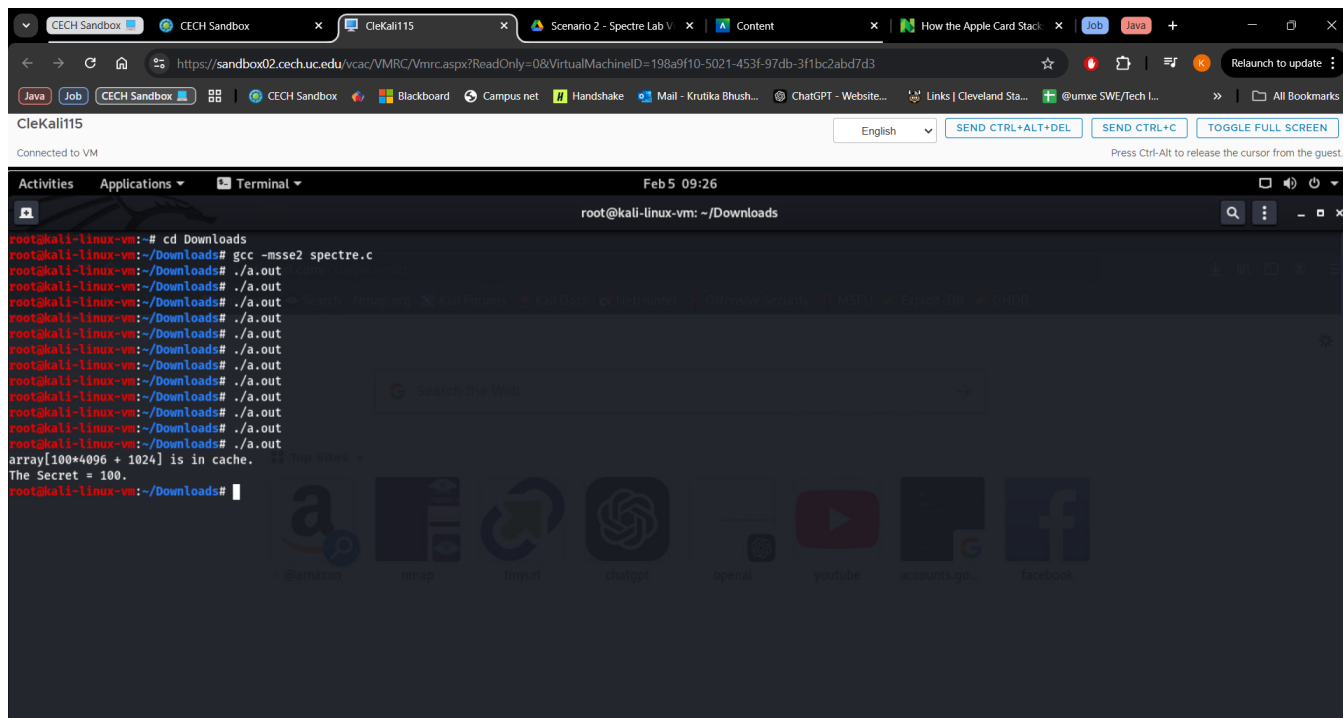
1. Provide the screenshot of the downloaded 'C' file.



2. Provide the screenshot of the compilation and execution of the 'c' code.



3. Provide the screenshot of the desired output showing the secret value.



Exercise 2: Lab_Port Scanning

Part One: Exploring Kali Linux

Look through the applications installed on your Kali Linux System.

1. List and identify any tools that you recognize.

- **Nmap (Network Mapper)** – A network scanning tool used for discovering hosts and services on a network, identifying open ports, and performing security audits.
- **Medusa** – A brute-force password-cracking tool that can perform dictionary attacks on various protocols, such as SSH, FTP, and HTTP.
- **Wireshark** – A network protocol analyzer used for packet capturing and network traffic analysis. It helps in troubleshooting network issues and detecting security threats.
- **Sqlmap** – An automated SQL injection tool used for testing and exploiting SQL injection vulnerabilities in web applications.

Open the terminal application, type “nmap” and look at the man page for nmap. Read through this page and answer the following questions:

2. According to the description, what tasks do system and network administrators use nmap for?

Nmap (Network Mapper) is a powerful tool used by system and network administrators for various tasks, including:

1. Network Discovery – Finding active hosts and devices in a network.
2. Port Scanning – Identifying open ports on a target system to determine services running.
3. Service and Version Detection – Determining the exact software running on open ports.
4. OS Detection – Identifying the operating system running on a target host.
5. Security Auditing – Detecting vulnerabilities, misconfigurations, and potential security risks.
6. Firewall Evasion and IDS Testing – Testing firewall rules and Intrusion Detection System (IDS) capabilities.
7. Performance Monitoring – Checking server uptime and response times.

3. What option would you use to treat all hosts as online (skip host discovery)?

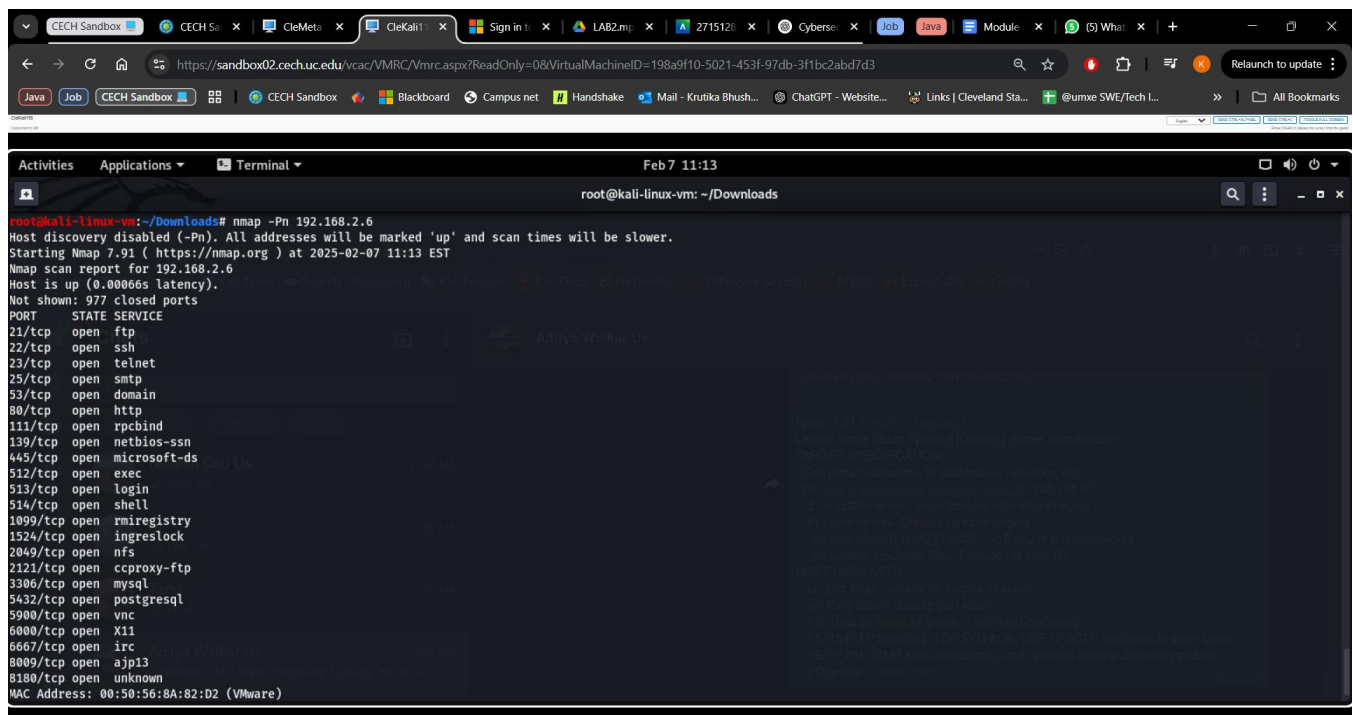
Option: -Pn

- By default, Nmap performs host discovery (pings the host before scanning).
- The -Pn option skips this step and treats all specified hosts as online.

Use case:

- Useful when scanning firewalled hosts that block ICMP ping requests.

nmap -Pn 192.168.2.6



```
root@kali-linux-vm: ~/Downloads
root@kali-linux-vm:~/Downloads# nmap -Pn 192.168.2.6
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times will be slower.
Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-07 11:13 EST
Nmap scan report for 192.168.2.6
Host is up (0.00066s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
145/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  mircregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:50:56:8A:82:D2 (VMware)
```

4. What option would you use to specify specific ports to scan?

Option: -p <port ranges>

- This allows scanning specific ports rather than all 65535 ports.
- You can specify:
 - A single port: -p 22
 - A range of ports: -p 20-80
 - Multiple ports: -p 22,80,443

- UDP, TCP, and SCTP separately:

- UDP: U:53
- TCP: T:21-25,80
- SCTP: S:9

Use case:

- When you only need to check if a particular service is running.

`nmap -p 80,443 192.168.2.6`

```
root@kali-linux-vm:~/Downloads# nmap -p 80,443 192.168.2.6
Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-07 11:15 EST
Nmap scan report for 192.168.2.6
Host is up (0.00081s latency).

PORT      STATE SERVICE
80/tcp    open  http
443/tcp    closed https
MAC Address: 00:50:56:8A:82:D2 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
root@kali-linux-vm:~/Downloads#
```

5. What option would you use to determine service and version info on open ports?

Option: `-sV`

- This probes open ports to identify running services and their versions.

Use case:

- Useful for penetration testing to check if outdated or vulnerable versions of software are running.

`nmap -sV 192.168.2.6`

```
root@kali-linux-vm: ~/Downloads
root@kali-linux-vm:~/Downloads# nmap -sV 192.168.2.6
Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-07 11:18 EST
Nmap scan report for 192.168.2.6
Host is up (0.00075s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #100000)
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi      GNU Classpath gmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:50:56:8A:82:D2 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

6. What option would you use to enable OS detection?

Option: -O

- This performs operating system detection using TCP/IP stack fingerprinting.

Use case:

- Helps in network security audits by identifying OS vulnerabilities.

nmap -O 192.168.2.6

```
root@kali-linux-vm:~/Downloads# nmap -O 192.168.2.6
Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-07 11:26 EST
Nmap scan report for 192.168.2.6
Host is up (0.00059s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:50:56:BA:82:D2 (VMware)
Device type: general purpose

Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.8 - 2.6.30
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.82 seconds
root@kali-linux-vm:~/Downloads#
```

Check to see if you Kali system is up to date. If not, install all updates.

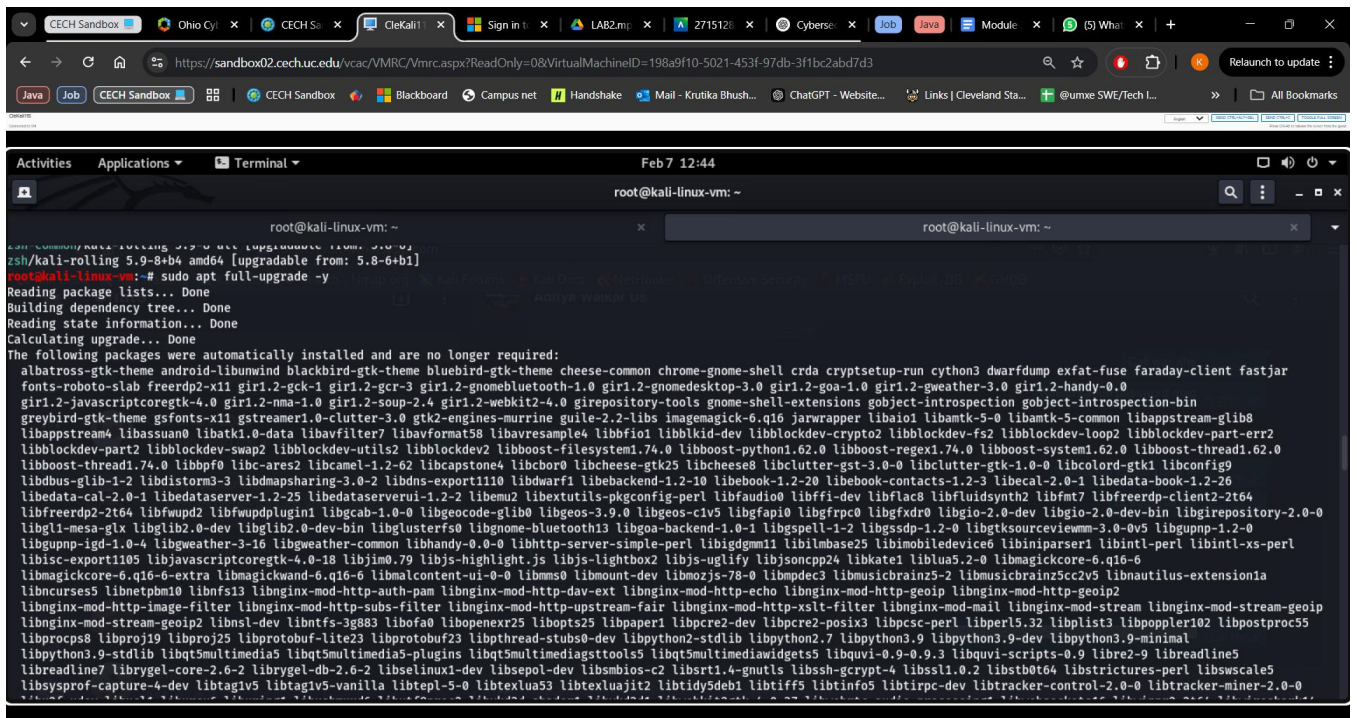
7. Paste a screen shot after updates are installed (or confirmed system is up to date.)

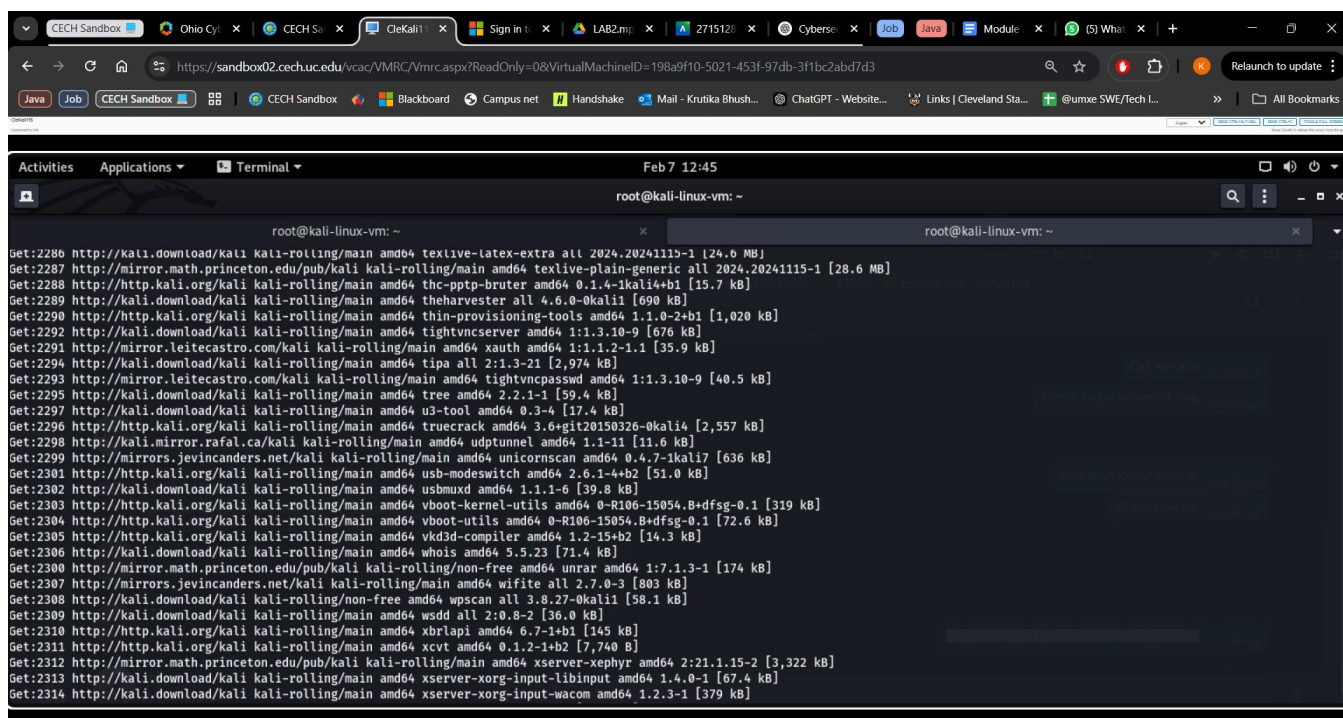
sudo apt update && sudo apt list --upgradable

This command checks for updates

```
root@kali-linux-vm: ~  
root@kali-linux-vm:~# sudo apt update && sudo apt list --upgradable  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
2266 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Listing... Done  
0trace/kali-rolling 0.01-3kali5 amd64 [upgradable from: 0.01-3kali2]  
aapt/kali-rolling 1:14-beta1-3 amd64 [upgradable from: 1:10.0.0+r36-3]  
accountsservice/kali-rolling 23.13-9-7 amd64 [upgradable from: 0.6.95-3]  
acl/kali-rolling 2.3.2-2+b1 amd64 [upgradable from: 2.2.53-10]  
adduser/kali-rolling 3.137 all [upgradable from: 3.118]  
adwaita-icon-theme/kali-rolling 47.0-2 all [upgradable from: 3.38.0-1]  
afflib-tools/kali-rolling 3.7.20-2+b1 amd64 [upgradable from: 3.7.19-1]  
aircrack-ng/kali-rolling 1:1.7+git20230807.4bf83fia-2 amd64 [upgradable from: 1:1.6+git2021030.91820bc-1]  
alsa-tools/kali-rolling 1.2.11-1.1 amd64 [upgradable from: 1.2.2-1]  
alsa-topology-conf/kali-rolling 1.2.5.1-3 all [upgradable from: 1.2.4-1]  
alsa-ucm-conf/kali-rolling 1.2.13-1 all [upgradable from: 1.2.4-2]  
amap/kali-rolling 5.4-4kali3 amd64 [upgradable from: 5.4-4kali2]  
anacron/kali-rolling 2.3-41 amd64 [upgradable from: 2.3-30]  
android-framework-res/kali-rolling 1:14-beta1-3 all [upgradable from: 1:10.0.0+r36-3]  
android-libaapt/kali-rolling 1:14-beta1-3 amd64 [upgradable from: 1:10.0.0+r36-3]  
android-libandroidfw/kali-rolling 1:14-beta1-3 amd64 [upgradable from: 1:10.0.0+r36-3]  
android-libbacktrace/kali-rolling 1:34.0-5-6 amd64 [upgradable from: 1:10.0.0+r36-7]  
android-libbase/kali-rolling 1:34.0-5-6 amd64 [upgradable from: 1:10.0.0+r36-7]  
android-libbrotli/kali-rolling 1:34.0-5-6 amd64 [upgradable from: 1:10.0.0+r36-7]  
android-liblog/kali-rolling 1:34.0-5-6 amd64 [upgradable from: 1:10.0.0+r36-7]  
android-libutils/kali-rolling 1:34.0-5-6 amd64 [upgradable from: 1:10.0.0+r36-7]  
android-libziparchive/kali-rolling 1:34.0-5-6 amd64 [upgradable from: 1:10.0.0+r36-7]
```

```
root@kali-linux-vm: ~  
root@kali-linux-vm:~# sudo apt list --upgradable  
cryptcat/kali-rolling 20031202-5kali7 amd64 [upgradable from: 20031202-5kali4]  
cryptsetup-bin/kali-rolling 2:2.7.5-1 amd64 [upgradable from: 2:2.3.5-1]  
cryptsetup-initramfs/kali-rolling 2:2.7.5-1 all [upgradable from: 2:2.3.5-1]  
cryptsetup/kali-rolling 2:2.7.5-1 amd64 [upgradable from: 2:2.3.5-1]  
cups-bsd/kali-rolling 2.4.10-2+b1 amd64 [upgradable from: 2.3.3op2-3+deb11u1]  
cups-client/kali-rolling 2.4.10-2+b1 amd64 [upgradable from: 2.3.3op2-3+deb11u1]  
cups-common/kali-rolling 2.4.10-2 all [upgradable from: 2.3.3op2-3+deb11u1]  
cups-pk-helper/kali-rolling 0.2.6-2.1 amd64 [upgradable from: 0.2.6-1+b1]  
curl/kali-rolling 8.11.1-1+b1 amd64 [upgradable from: 7.74.0-1.2]  
curlftpfs/kali-rolling 0.9.2-10 amd64 [upgradable from: 0.9.2-9+b1]  
cutycapt/kali-rolling 0.0+20130714-1 amd64 [upgradable from: 0.0-svn10-0.1+b2]  
cymothoa/kali-rolling 1-beta-1kali3 amd64 [upgradable from: 1-beta-1kali2]  
cython3/kali-rolling 3.0.11+dfsg-1+b1 amd64 [upgradable from: 0.29.21-3+b1]  
darkstat/kali-rolling 3.0.721-2 amd64 [upgradable from: 3.0.719-1+b1]  
dash/kali-rolling 0.5.12-12 amd64 [upgradable from: 0.5.11+git20200708+dd9ef66-5]  
davtest/kali-rolling 1.2+git20230307.34d31db-0kali1 all [upgradable from: 1.0-1kali4]  
dbd/kali-rolling 1.50-1kali7 amd64 [upgradable from: 1.50-1kali6]  
dbus-user-session/kali-rolling 1.16.0-1 amd64 [upgradable from: 1.12.20-2]  
dbus-x11/kali-rolling 1.16.0-1 amd64 [upgradable from: 1.12.20-2]  
dbus/kali-rolling 1.16.0-1 amd64 [upgradable from: 1.12.20-2]  
dc3dd/kali-rolling 7.3.1-3 amd64 [upgradable from: 7.2.646-4]  
dcfldd/kali-rolling 1.9.2-1 amd64 [upgradable from: 1.7-3]  
dconf-cli/kali-rolling 0.40.0-5 amd64 [upgradable from: 0.38.0-2]  
dconf-gsettings-backend/kali-rolling 0.40.0-5 amd64 [upgradable from: 0.38.0-2]  
dconf-service/kali-rolling 0.40.0-5 amd64 [upgradable from: 0.38.0-2]  
dcrw/kali-rolling 9.28-7 amd64 [upgradable from: 9.28-2]  
ddrescue/kali-rolling 1.99.13-0kali1 amd64 [upgradable from: 1.99.11-0kali4]  
debconf-i18n/kali-rolling 1.5.89 all [upgradable from: 1.5.75]
```

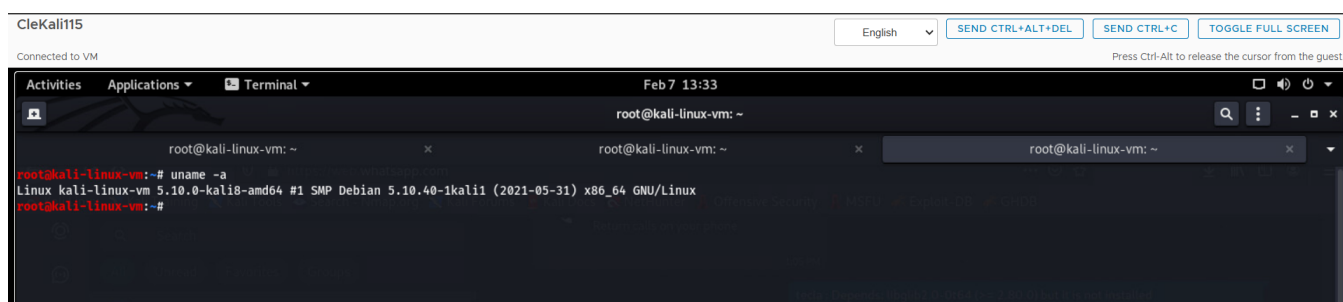





8. What distribution of Linux is Kali based on? (Hint: use the uname command)

Kali Linux is based on Debian.

uname -a command output includes "Debian", which confirms that Kali is a Debian-based distribution. It follows a rolling release model and is specifically tailored for penetration testing, cybersecurity, and digital forensics.



Part Two: Starting Metasploit Framework

Use the Metasploit Unleashed site for help completing this portion of the lab.

9. In your own words, explain what the Metasploit Framework is.

The Metasploit Framework is an open-source cybersecurity tool used for penetration testing and security assessments. It allows ethical hackers and security professionals to simulate real-world cyberattacks to identify and fix vulnerabilities in networks, applications, and devices. With a vast collection of exploits, payloads, and tools, Metasploit helps automate attacks and test defenses, ensuring that security weaknesses are addressed before malicious hackers can exploit them. It is widely used in the cybersecurity field to improve overall system security and resilience.

Start the postgresql, make sure the 'msfdb' is initialized, and open the msfconsole.

In Terminal type: services postgresql start

In Terminal type: Msdb init

In Terminal type: msfconconsole

10. Paste as screen show of each of the commands used to complete these steps

Start the postgresql and check status

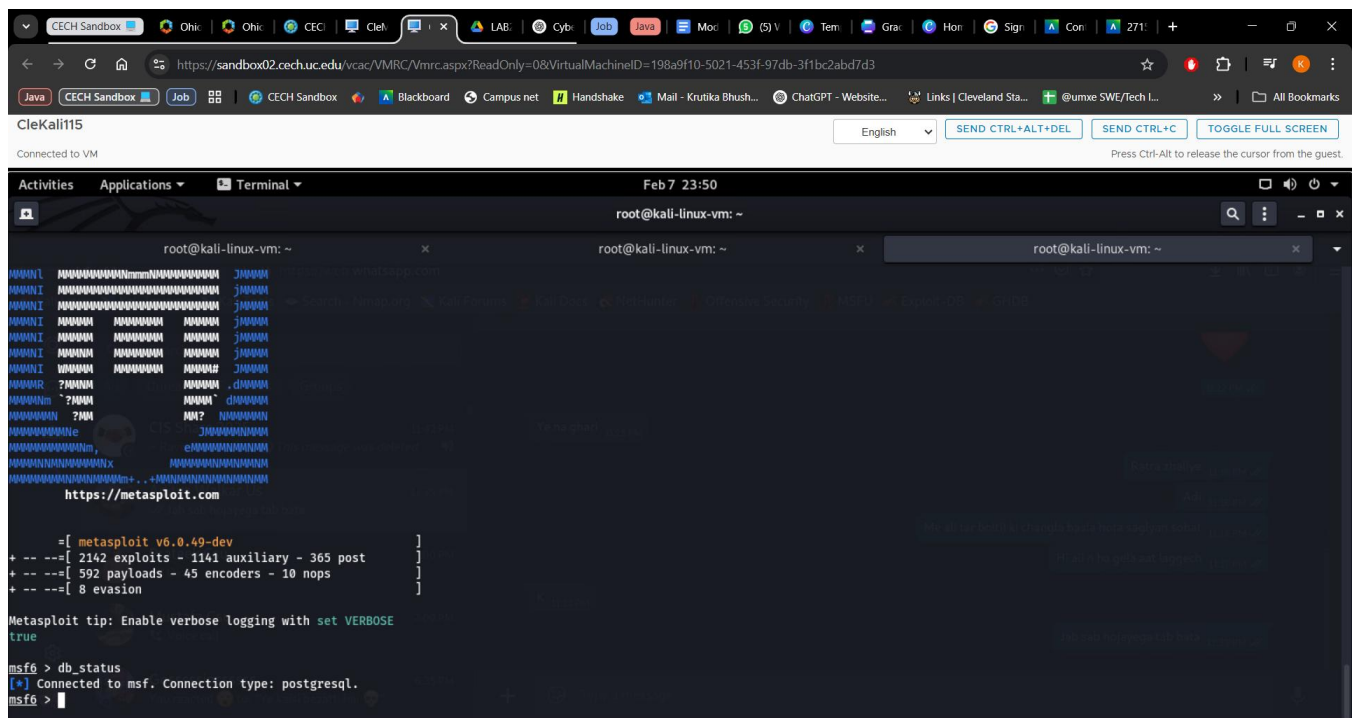
service postgresql start

service postgresql status

Initialize msfdb and check status

msfdb init

msfdb status



11. What is the IP and subnet mask of your Kali Linux System?

The IP address of my Kali Linux system is **192.168.2.7**, and the subnet mask is **255.255.255.0**.

12. What is the Network Address of your network?

The network address of my network can be determined using the IP address and subnet mask. Since my IP is **192.168.2.7** and the subnet mask is **255.255.255.0**, the network address is **192.168.2.0**.

```

msf6 > ifconfig
[*] exec: ifconfig

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.2.7 netmask 255.255.255.0 broadcast 192.168.2.255
    inet6 fe80::250:56ff:fe8a:6cae prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:8a:6c:ae txqueuelen 1000 (Ethernet)
    RX packets 832065 bytes 4138083411 (3.8 GiB)
    RX errors 0 dropped 28 overruns 0 frame 0
    TX packets 721243 bytes 59896842 (57.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 13490140 bytes 3183958312 (2.9 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13490140 bytes 3183958312 (2.9 GiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

13. What hosts are listed in your database now? (Show a screen shot)

```

msf6 > hosts

Hosts
=====

address      mac      name      os_name      os_flavor      os_sp      purpose      info      comments
-----
192.168.2.3   WIN-678LK3JBN3Q  Windows 7  Enterprise  SP1        client
192.168.2.6

```

14. What services are listed in your database now? (Show a screen shot)

```

msf6 > services

Services
=====

host      port      proto      name      state      info
----
192.168.2.3  445      tcp

```

Run an nmap ping sweep in your network. (nmap -sn 192.168.2.0/24)

15. Paste a screen show of the command and results.

```
msf6 > nmap -sn 192.168.2.6
[*] exec: nmap -sn 192.168.2.6

Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-08 00:20 EST
Nmap scan report for 192.168.2.6
Host is up (0.0018s latency).
MAC Address: 00:50:56:8A:82:D2 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
msf6 > |
```

16. What is the IP address of you metasploit2 target? (Take note of this for future assignments)

The IP address for my metasploit2 target is 192.168.2.6

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:56:8a:82:d2
          inet addr:192.168.2.6  Bcast:192.168.2.255  Mask:255.255.255.0
          inet6 addr: fe80::250:56ff:fe8a:82d2/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:123778 errors:10 dropped:11 overruns:0 frame:0
          TX packets:107553 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:13970593 (13.3 MB)  TX bytes:25250358 (24.0 MB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:99427 errors:0 dropped:0 overruns:0 frame:0
          TX packets:99427 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:49623921 (47.3 MB)  TX bytes:49623921 (47.3 MB)
```

Run a nmap scan against your target. This time use db_nmap to store the results in your database. Specify options to meet the following criteria:

- Scan ports 22,53,80,443 and 55432
- Run OS detection

17. Show a screen show if the command and the results.

db_nmap → Saves results in the Metasploit database.

-p 22,53,80,443,55432 → Scans the specified ports.

-O → Enables OS detection to identify the target system's operating system.

```
msf6 > db_nmap -p 22,53,80,443,55432 -O 192.168.2.6
[*] Nmap: Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-08 00:37 EST
[*] Nmap: Nmap scan report for 192.168.2.6
[*] Nmap: Host is up (0.00093s latency).
[*] Nmap: PORT      STATE SERVICE
[*] Nmap: 22/tcp    open  ssh
[*] Nmap: 53/tcp    open  domain
[*] Nmap: 80/tcp    open  http
[*] Nmap: 443/tcp   closed https
[*] Nmap: 55432/tcp closed unknown
[*] Nmap: MAC Address: 00:50:56:8A:82:D2 (VMware)
[*] Nmap: Device type: general purpose
[*] Nmap: Running: Linux 2.6.X
[*] Nmap: OS CPE: cpe:/o:linux:linux_kernel:2.6
[*] Nmap: OS details: Linux 2.6.8 - 2.6.30
[*] Nmap: Network Distance: 1 hop
[*] Nmap: OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 1.74 seconds
```

18. What services are now listed in your database? (Show a screen shot) (Type: hosts)

```
msf6 > hosts

Hosts
=====

address      mac          name          os_name      os_flavor    os_sp  purpose  info  comments
-----
192.168.2.3   Aditya Vaidya WIN-678LK3JBN3Q Windows 7    Enterprise   SP1    client
192.168.2.6   00:50:56:8a:82:d2 Linux         Linux        2.6.X       server

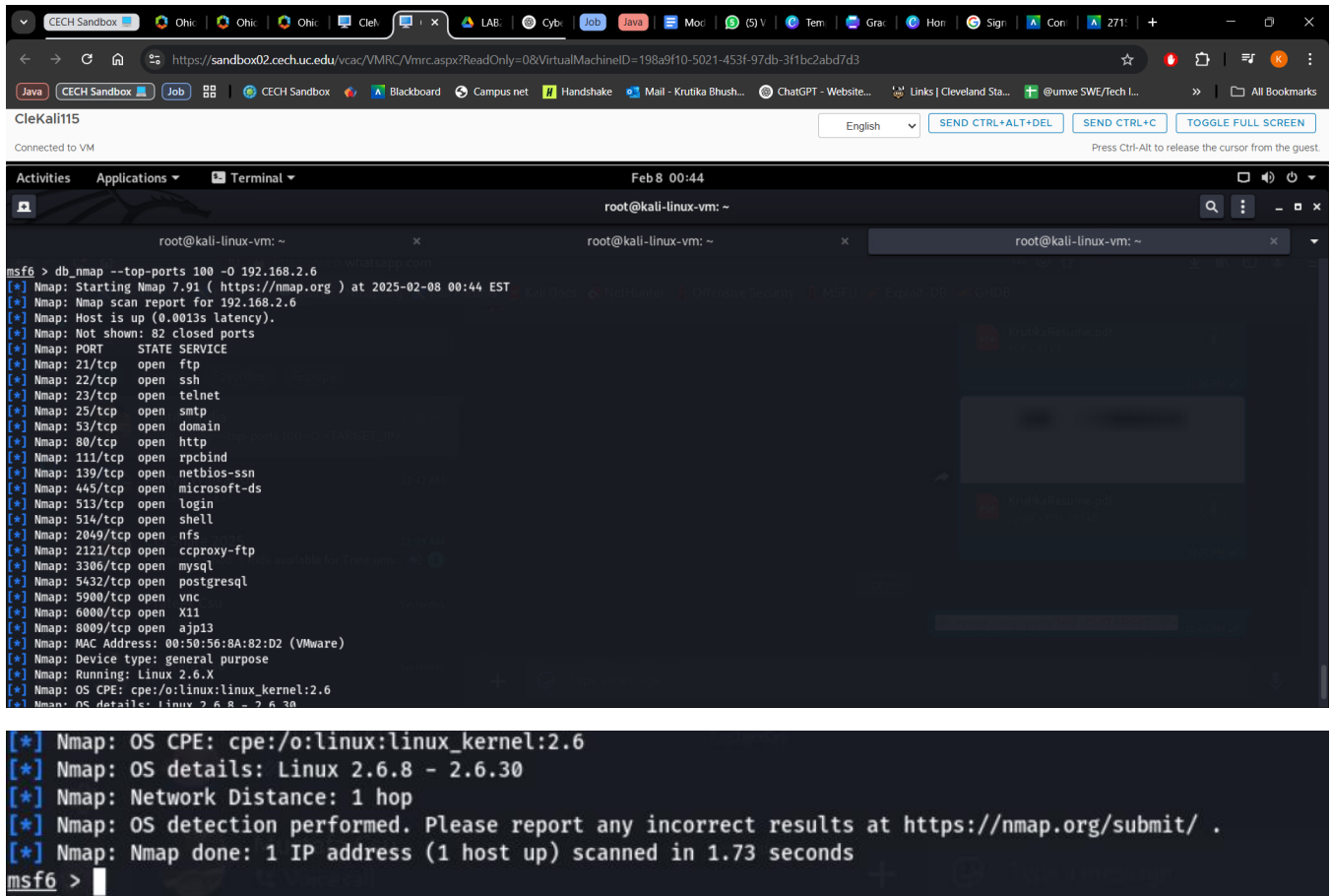
msf6 > services

Services
=====

host      port  proto  name    state  info
-----
192.168.2.3 445   tcp    ssh     open
192.168.2.6 22    tcp    ssh     open
192.168.2.6 53    tcp    domain open
192.168.2.6 80    tcp    http   open
192.168.2.6 443   tcp    https  closed
192.168.2.6 55432 tcp    https  closed

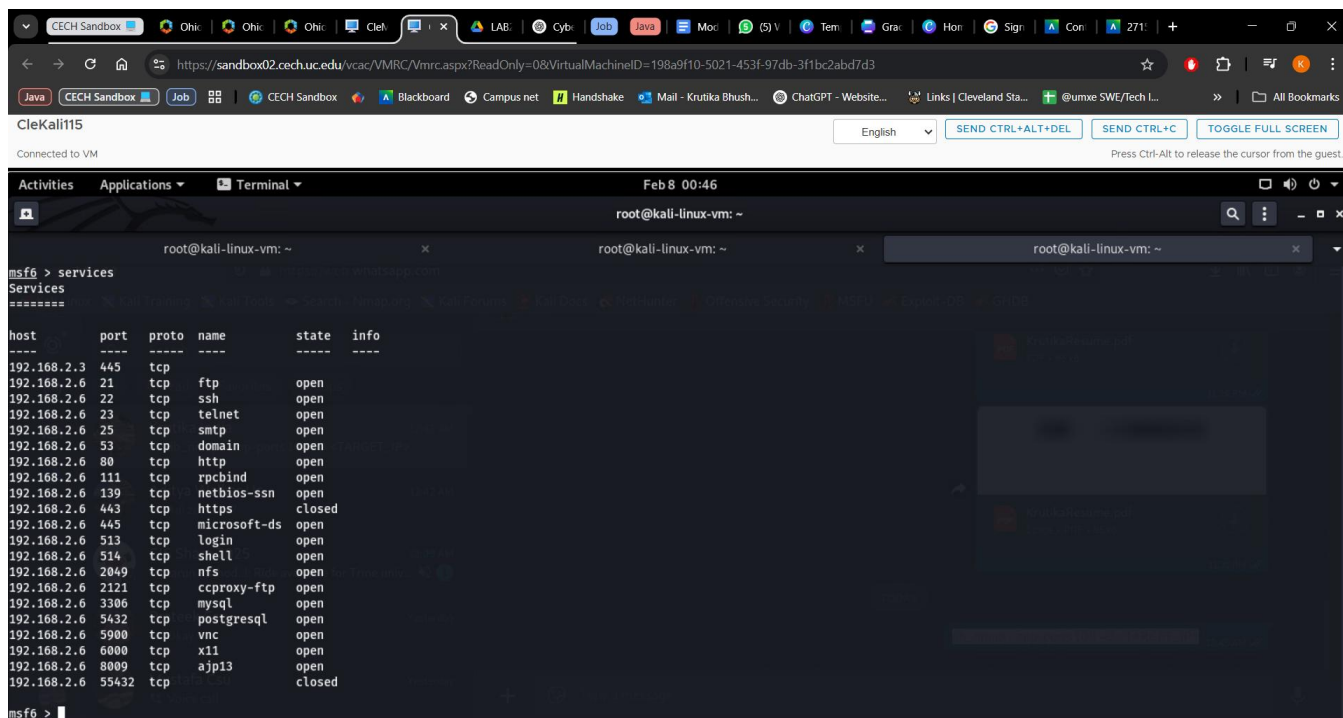
msf6 > 
```

19. Run another nmap scan against your target, this time choose the top 100 ports. (Show a screen shot)



```
msf6 > db_nmap --top-ports 100 -O 192.168.2.6
[*] Nmap: Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-08 00:44 EST
[*] Nmap: Nmap scan report for 192.168.2.6
[*] Nmap: Host is up (0.0013s latency).
[*] Nmap: Not shown: 82 closed ports
[*] Nmap: PORT      STATE SERVICE
[*] Nmap: 21/tcp    open  ftp
[*] Nmap: 22/tcp    open  ssh
[*] Nmap: 23/tcp    open  telnet
[*] Nmap: 25/tcp    open  smtp
[*] Nmap: 53/tcp    open  domain
[*] Nmap: 80/tcp    open  http
[*] Nmap: 111/tcp   open  rpcbind
[*] Nmap: 139/tcp   open  netbios-ssn
[*] Nmap: 445/tcp   open  microsoft-ds
[*] Nmap: 513/tcp   open  login
[*] Nmap: 514/tcp   open  shell
[*] Nmap: 2049/tcp  open  nfs
[*] Nmap: 2121/tcp  open  cproxy-ftp
[*] Nmap: 3306/tcp  open  mysql
[*] Nmap: 5432/tcp  open  postgresql
[*] Nmap: 5900/tcp  open  vnc
[*] Nmap: 6000/tcp  open  x11
[*] Nmap: 8009/tcp  open  ajp13
[*] Nmap: MAC Address: 00:50:56:8A:82:D2 (VMware)
[*] Nmap: Device type: general purpose
[*] Nmap: Running: Linux 2.6.X
[*] Nmap: OS CPE: cpe:/o:linux:linux_kernel:2.6
[*] Nmap: OS details: Linux 2.6.8 - 2.6.30
[*] Nmap: OS CPE: cpe:/o:linux:linux_kernel:2.6
[*] Nmap: OS details: Linux 2.6.8 - 2.6.30
[*] Nmap: Network Distance: 1 hop
[*] Nmap: OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 1.73 seconds
msf6 >
```

20. What services are running on your target? (Show a screen shot)



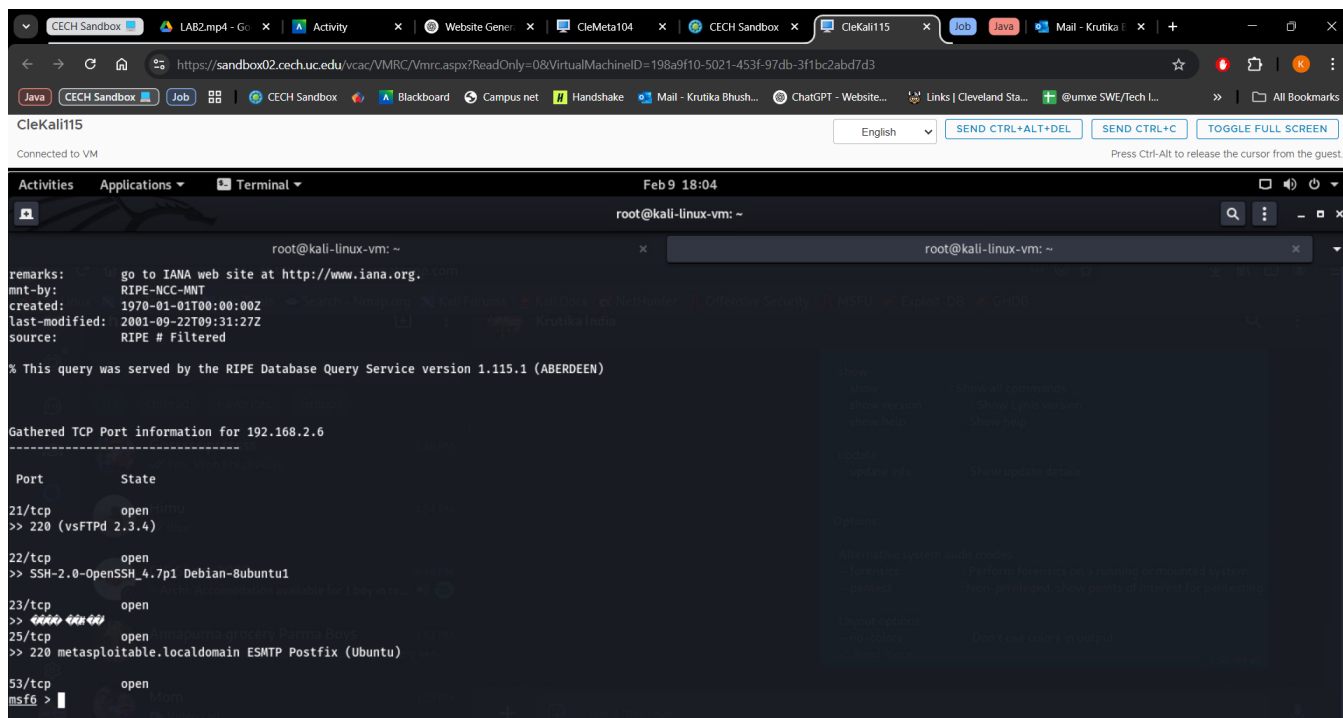
Run a scan with one of Metasploits built in port scanning tools against your target. You choose the tool and the options.

21. Paste a screen shot of the options you choose and the results of running the scanner

Dmitry -winsepfb 192.168.2.6

This command runs Dmitry (Deepmagic Information Gathering Tool) with multiple options enabled against the target IP 192.168.2.6.

Option	Description
-w	Perform a whois lookup on the target (both domain and IP-based).
-i	Retrieve IP address information for the target.
-n	Perform an NS lookup (Name Server lookup) to gather DNS details.
-s	Conduct a subdomain search to find related domains.
-e	Perform an email address search on the target domain.
-p	Scan open ports on the target system.
-f	Enable full detailed output , providing an extensive report.



22. Did you find any additional services that are running? List the service and ports below.

Based on the scan results, these additional services were found running on **192.168.2.6**:

Port	Protocol	Service	State
21	TCP	FTP	Open
22	TCP	SSH	Open
23	TCP	Telnet	Open
25	TCP	SMTP	Open
53	TCP	DNS	Open
80	TCP	HTTP	Open
111	TCP	RPCBind	Open
139	TCP	NetBIOS-SSN	Open
445	TCP	Microsoft-DS	Open

Port	Protocol	Service	State
513	TCP	Login	Open
514	TCP	Shell	Open
2049	TCP	NFS	Open
2121	TCP	CCProxy-FTP	Open
3306	TCP	MySQL	Open
5432	TCP	PostgreSQL	Open
5900	TCP	VNC	Open
6000	TCP	X11	Open
8009	TCP	AJP13	Open

Closed Ports:

- 443 (HTTPS)
- 55432 (Unknown Service)

The screenshot displays a Kali Linux desktop environment. At the top, a web browser window is open, showing a URL that appears to be a virtual machine configuration page. Below the browser, a terminal window is active, displaying a list of open and closed ports for a host at 192.168.2.6. The terminal output is as follows:

```

Services
=====
host      port  proto name      state  info
-----
192.168.2.3 445   tcp    ftp       open
192.168.2.6 21    tcp    ftp       open
192.168.2.6 22    tcp    ssh       open
192.168.2.6 23    tcp    telnet    open
192.168.2.6 25    tcp    smtp      open
192.168.2.6 53    tcp    domain    open
192.168.2.6 80    tcp    http      open
192.168.2.6 111   tcp    rpcbind   open
192.168.2.6 139   tcp    netbios-ssn open
192.168.2.6 443   tcp    https     closed
192.168.2.6 445   tcp    microsoft-ds open
192.168.2.6 513   tcp    login     open
192.168.2.6 514   tcp    shell     open
192.168.2.6 2049  tcp    nfs       open
192.168.2.6 2121  tcp    ccproxy-ftp open
192.168.2.6 3306  tcp    mysql     open
192.168.2.6 5432  tcp    postgresql open
192.168.2.6 5900  tcp    vnc       open
192.168.2.6 6000  tcp    x11       open
192.168.2.6 8009  tcp    ajp13     open
192.168.2.6 55432 tcp     closed

```

The terminal window also shows the prompt 'msf6 >' at the bottom, indicating that the user is in the Metasploit framework console.

Exit the msfconsole.

Part Four: Bonus

Choose one of the other port scanning tools discussed in the course materials (p0f, Xprobe2, Masscan, Netcat). Run any kind of scan against your target system that you like. Then record a short video (under 3 minutes) and explain what options you choose and describe the results.

Attached Separately in the submission

Exercise 3: Lab_Cryptography (Cryptography, Encoding, and Numbering Systems)

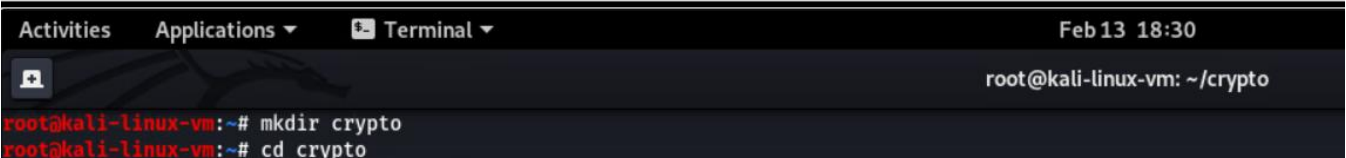
Part Zero: Hashing

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- `mkdir crypto`
 - `cd crypto`
-

CleKali115

Connected to VM

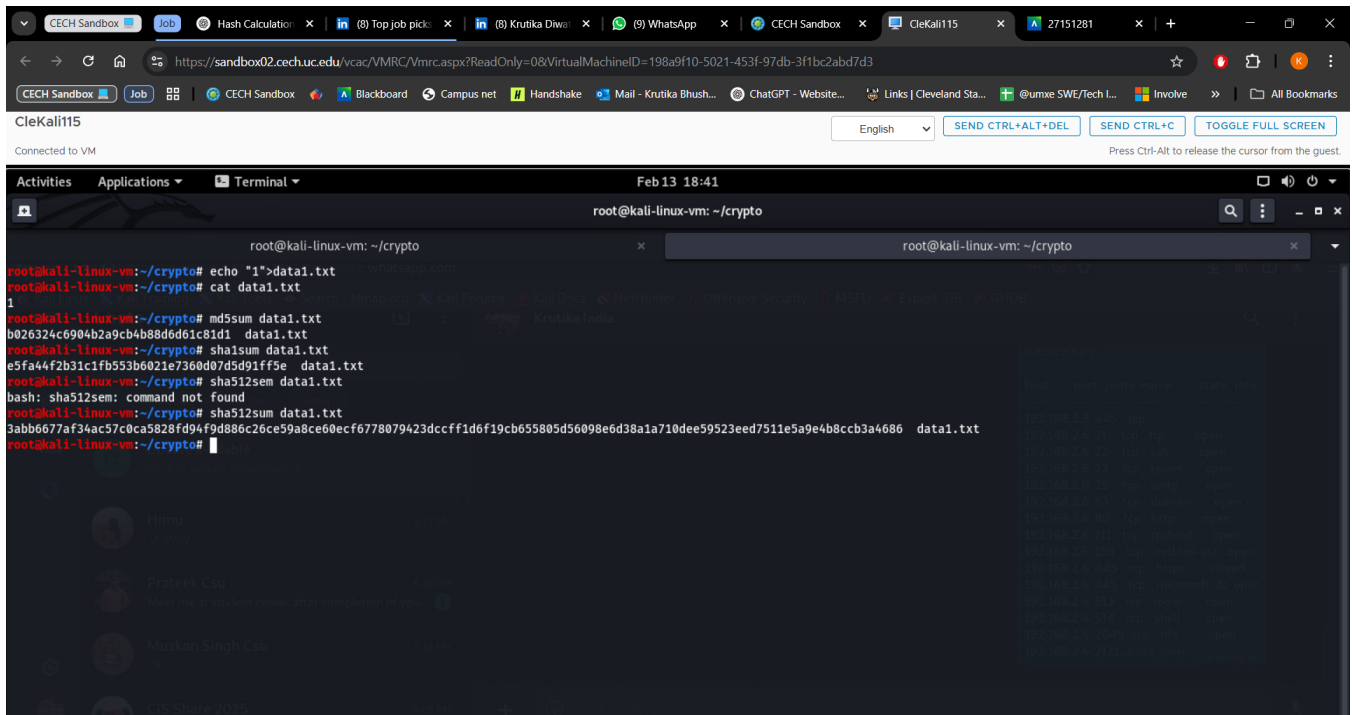


```
Activities Applications Terminal Feb 13 18:30
root@kali-linux-vm: ~/crypto
root@kali-linux-vm:~# mkdir crypto
root@kali-linux-vm:~# cd crypto
```

Create a file, calculate hashes using different algorithms. Compare the output hash values:

- `echo "1" > data1.txt`
- `cat data.txt`
- `md5sum data1.txt`
- `sha1sum data1.txt`

- sha512sum data1.txt



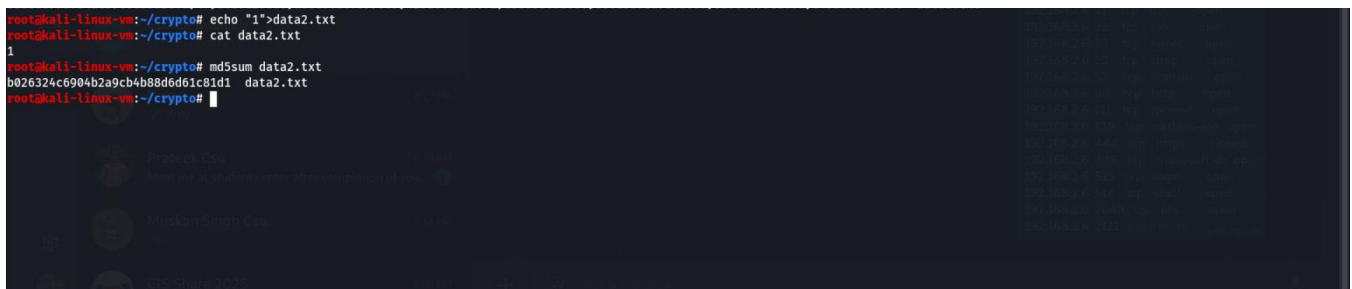
```
root@kali-linux-vm: ~/crypto
root@kali-linux-vm:~/crypto# echo "1">data1.txt
root@kali-linux-vm:~/crypto# cat data1.txt
1
root@kali-linux-vm:~/crypto# md5sum data1.txt
b026324c6904b2a9cb4b88d6d61c81d1  data1.txt
root@kali-linux-vm:~/crypto# sha1sum data1.txt
e5fa44f2b3c1fb553b6021e7360d07d5d91ff5e  data1.txt
root@kali-linux-vm:~/crypto# sha512sum data1.txt
bash: sha512sum: command not found
root@kali-linux-vm:~/crypto# sha512sum data1.txt
3abb6677af34ac57c0ca5828fd94f9d886c26ce59a8ce60ecf6778079423dcccfd6f19cb655805d56098e6d38a1a710dee59523eed7511e5a9e4b8ccb3a4686  data1.txt
root@kali-linux-vm:~/crypto#
```

Create another file with the same content. Note that the filename does not change the hash value, but the file content does change:

- echo "1" > data2.txt

- cat data2.txt

- md5sum data2.txt



```
root@kali-linux-vm:~/crypto# echo "1">data2.txt
root@kali-linux-vm:~/crypto# cat data2.txt
1
root@kali-linux-vm:~/crypto# md5sum data2.txt
b026324c6904b2a9cb4b88d6d61c81d1  data2.txt
root@kali-linux-vm:~/crypto#
```

Create a third file with different content. Note that you can copy and paste thousands of words in the third file. Calculate hash values, and note that hash length does not change depending on the input length, but the hash values change.

o `echo "lots of characters here" > data3.txt`

o `cat data3.txt`

- `md5sum data3.txt`
- `sha1sum data3.txt mk`
- `sha512sum data3.txt`

```
root@kali-linux-vm:~/crypto# echo "lots of characters here">data3.txt
root@kali-linux-vm:~/crypto# cat data3.txt
lots of characters here
root@kali-linux-vm:~/crypto# md5sum data3.txt
a0eeabdd87e294c69d1d58415eeb438  data3.txt
root@kali-linux-vm:~/crypto# sha1sum data3.txt
77c953c311a8d286d196ee83c873fc3046a64f91  data3.txt
root@kali-linux-vm:~/crypto# sha512sum data3.txt
bf14b9ba7056f4e1466376e48035b02d44dca1fe47f8847fa42d48dd63235f8d86f1d24dc65f5bf3a51c5ceaa1325091eae6163e8ceb8d60bfe4be0678504091  data3.txt
```

Compare the hash values of created files all at once using the md5deep tool. Note that you may need to calculate many files you collected from the suspected machine as a forensics investigator.

- `md5deep data*.txt`

```
root@kali-linux-vm:~/crypto# md5deep data*.txt
b026324c6904b2a9cb4b88d6d61c81d1  /root/crypto/data1.txt
b026324c6904b2a9cb4b88d6d61c81d1  /root/crypto/data2.txt
a0eeabdd87e294c69d1d58415eeb438  /root/crypto/data3.txt
root@kali-linux-vm:~/crypto# █
```

Module Assessment

Question 1. Will the hash length change depending upon the characters of text to be encrypted ? If no, prove it with the help of screenshots .

No, the hash length does not change depending on the characters of the text being encrypted. Hash functions produce a fixed-length output regardless of the input size or content.

For example:

- **SHA-256** always produces a 256-bit hash (64 hexadecimal characters).
- **MD5** always produces a 128-bit hash (32 hexadecimal characters).

Each command will display a hash of 64 hexadecimal characters, regardless of the input text, proving that the hash length does not depend on the input content.

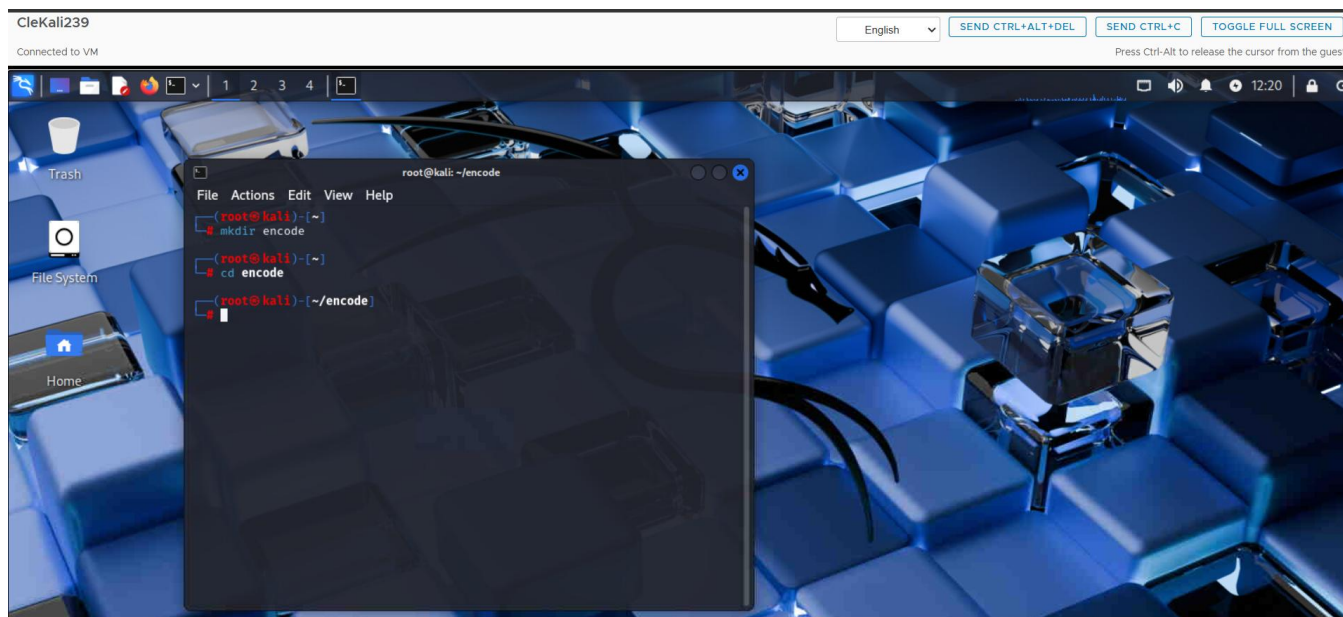
```
root@kali-linux-vm:~/crypto# echo -n "Hello, World!" | openssl dgst -sha256
echo -n "OpenAI" | openssl dgst -sha256
echo -n "Cryptography is fun!" | openssl dgst -sha256
echo -n "12345" | openssl dgst -sha256
echo -n "Linux Rocks!" | openssl dgst -sha256
(stdin)= dffd6021bb2bd5b0af676290809ec3a53191dd81c7f70a4b28688a362182986f
(stdin)= 8b7d1a3187ab355dc31bc683aaa71ab5ed217940c12196a9cd5f4ca984babfa4
(stdin)= 38def7e46a4674d78762823df2cb7dcddfb7d69692efdf91e11a27e540e92a57
(stdin)= 5994471abb01112afcc18159f6cc74b4f511b99806da59b3caf5a9c173cacfc5
(stdin)= 0042f983e187049499f83a1742aabb9d4ffc63a4a1c913566b8ae8eda92512ff
root@kali-linux-vm:~/crypto#
```

Module Activity Description

Part One: Encoding & Decoding

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

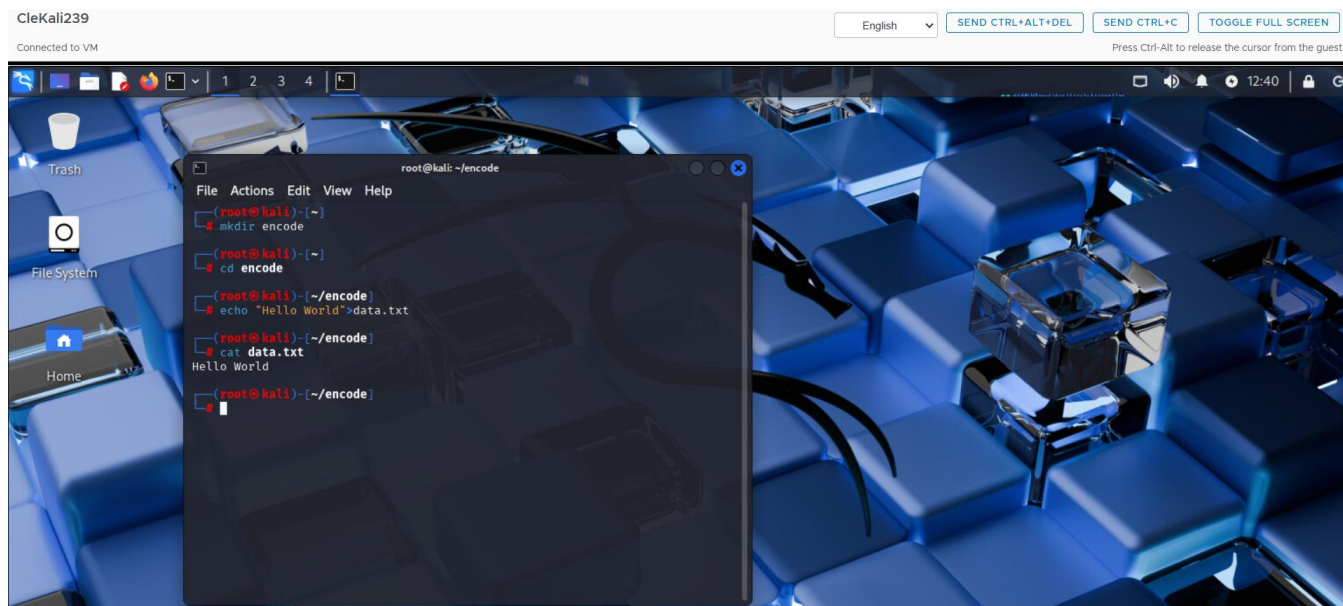
- **mkdir encode**
 - **cd encode**
-



Create a file and check the content:

- **echo "Hello World!" > data.txt**

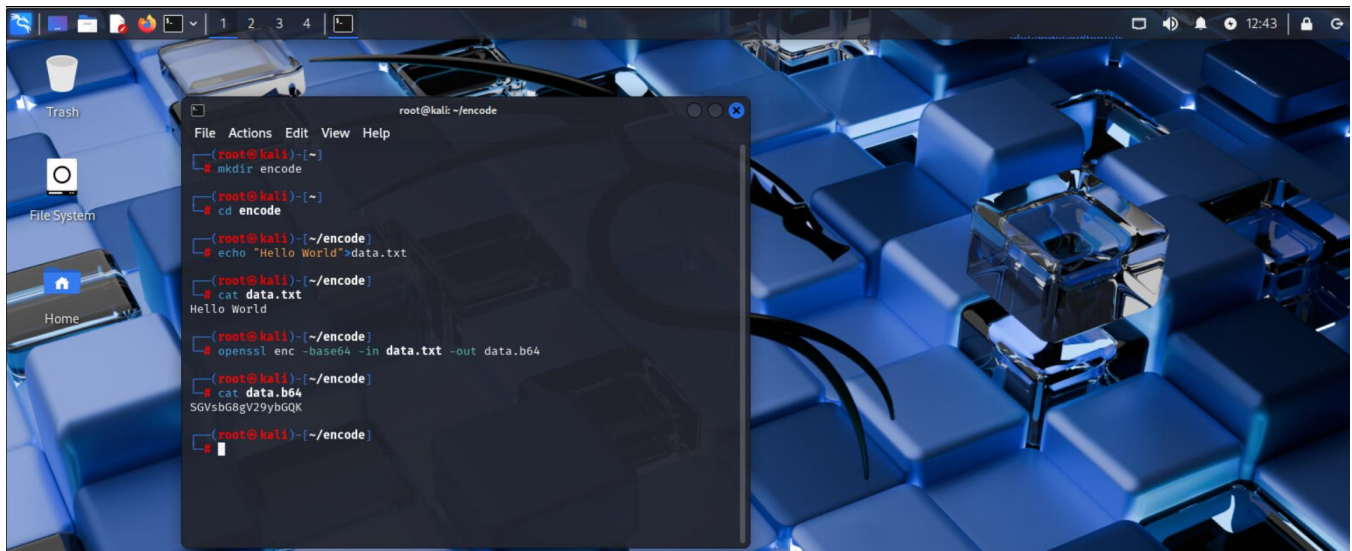
- **cat data.txt**



Perform Base64 encoding of the file using openssl utility and check the content:

- **openssl enc -base64 -in data.txt -out data.b64**

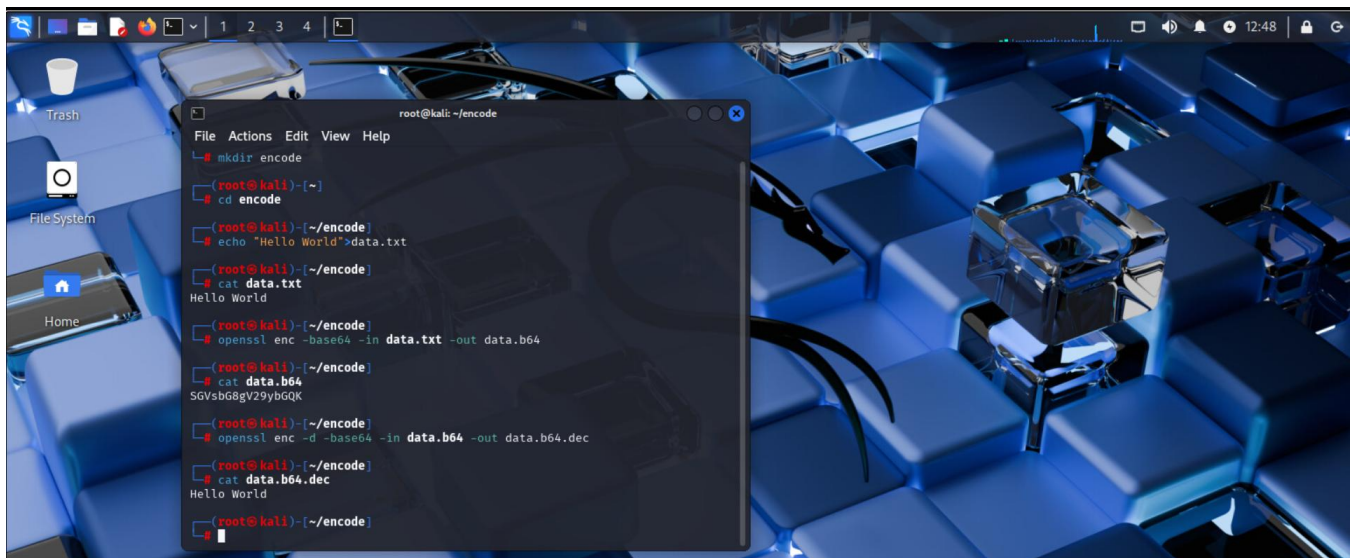
- **cat data.b64**



Decode the encoded file using openssl utility and check the content:

- **openssl enc -d -base64 -in data.b64 -out data.b64.dec**

- **cat data.b64.dec**



Module Assessment

Question 2. List the files from encode folder and compare the md5 hashes.
What is the conclusion from comparing md5sum with the directory.

```
(root@kali)-[~/encode]
# ls
data.b64  data.b64.dec  data.txt

(root@kali)-[~/encode]
# md5sum data.txt
md5sum data.b64
md5sum data.b64.dec
e59ff97941044f85df5297e1c302d260  data.txt
3558e8202f9eda6f06c561e617be8af3  data.b64
e59ff97941044f85df5297e1c302d260  data.b64.dec

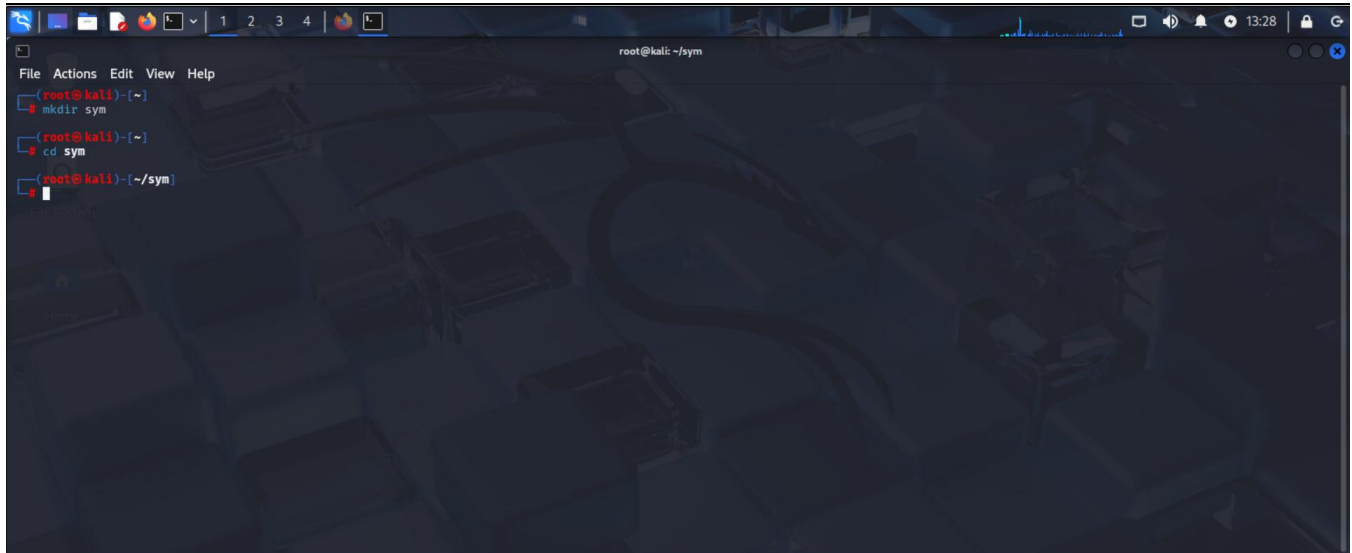
(root@kali)-[~/encode]
#
```

Module Activity Description: Part Two: Symmetric Encryption

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- **mkdir sym**

- **cd sym**



A terminal window screenshot showing the execution of two commands. The prompt is root@kali: ~. The first command is mkdir sym, and the second is cd sym. The prompt changes to root@kali: ~/sym after the second command. The terminal has a dark background with a faint cityscape pattern.

```
root@kali: ~  
- mkdir sym  
root@kali: ~  
- cd sym  
root@kali: ~/sym
```

Create a file and check the content:

- **echo "Hello World!" > data.txt**

- **cat data.txt**

Encrypt (AES encryption) the file using openssl utility and check the content:

- **openssl enc -aes-256-cbc -in data.txt -out data.aes**

Openssl will ask for a password to protect the key. Type a password.

- **cat data.aes**

```
root@kali: ~/sym
File Actions Edit View Help
root@kali)~)
# cd sym
root@kali)~/sym)
# echo "Hello World1">data.txt
root@kali)~/sym)
# cat data.txt
Hello World1
root@kali)~/sym)
# openssl enc -aes-256-cbc -in data.txt -out data.aes
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
root@kali)~/sym)
# cat data.aes
Salted__*\q***m+!GrV!+}**
root@kali)~/sym)
#
```

Decrypt the encrypted file using openssl utility and check the content:

- **openssl enc -d -aes-256-cbc -in data.aes -out data.aes.dec**

- **cat data.aes.dec**

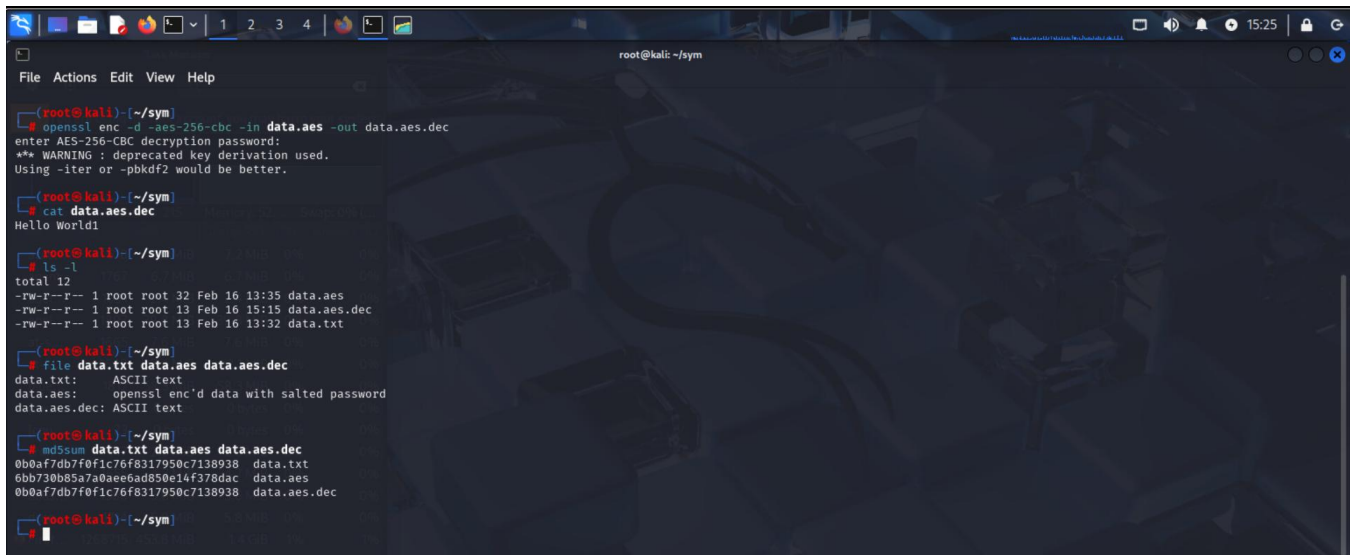
```
root@kali: ~/sym
File Actions Edit View Help
root@kali)~)
# cd sym
root@kali)~/sym)
# echo "Hello World1">data.txt
root@kali)~/sym)
# cat data.txt
Hello World1
root@kali)~/sym)
# openssl enc -aes-256-cbc -in data.txt -out data.aes
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
root@kali)~/sym)
# cat data.aes
Salted__*\q***m+!GrV!+}**
root@kali)~/sym)
# openssl enc -d -aes-256-cbc -in data.aes -out data.aes.dec
enter AES-256-CBC decryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
root@kali)~/sym)
# cat data.aes.dec
Hello World1
```

Module Assessment

Question 3 - List the files, see the filetypes, and compare the md5 hashes. What utility was used for AES encryption and explain it?

Utility: openssl

- openssl enc: Encryption command
- -aes-256-cbc: AES encryption with a 256-bit key in CBC mode
- AES (Advanced Encryption Standard) is a symmetric encryption algorithm widely used for securing data.
- CBC (Cipher Block Chaining) adds randomness by XORing each plaintext block with the previous ciphertext block before encryption.



```
(root@kali) ~/sym
$ openssl enc -d -aes-256-cbc -in data.aes -out data.aes.dec
enter AES-256-CBC decryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(root@kali) ~/sym
$ cat data.aes.dec
Hello World!

(root@kali) ~/sym
$ ls -l
total 12
-rw-r--r-- 1 root root 32 Feb 16 13:35 data.aes
-rw-r--r-- 1 root root 13 Feb 16 15:15 data.aes.dec
-rw-r--r-- 1 root root 13 Feb 16 13:32 data.txt

(root@kali) ~/sym
$ file data.txt data.aes data.aes.dec
data.txt:      ASCII text
data.aes:      openssl enc'd data with salted password
data.aes.dec:  ASCII text

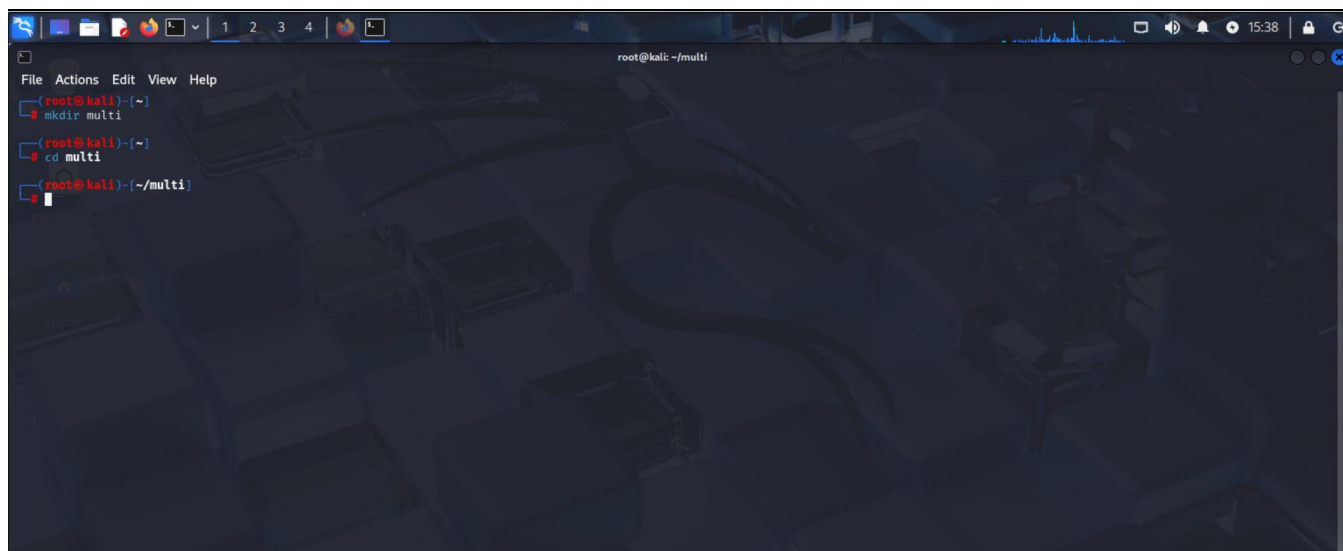
(root@kali) ~/sym
$ md5sum data.txt data.aes data.aes.dec
0bbaf7db7f0f1c76f8317950c7138938  data.txt
0bb730b85a7a0aee6ad80e14f378dac  data.aes
0bbaf7db7f0f1c76f8317950c7138938  data.aes.dec

(root@kali) ~/sym
```

Module Activity Description: Part Three: Multiple Operations

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- mkdir multi
 - cd multi
-

A terminal window on a Kali Linux system. The window title is 'root@kali: ~/multi'. The terminal shows the following commands and output:

```
root@kali:~# mkdir multi
root@kali:~# cd multi
root@kali:~/multi#
```

The background of the terminal has a dark, abstract, geometric pattern.

Create a file and check the content:

- **echo "Hello World!" > data.txt**
- **cat data.txt**

First encrypt the file (AES encryption) and then encode it (Base64) using openssl and check the content:

- **openssl enc -aes-256-cbc -base64 -in data.txt -out data.aes.b64**

OpenSSL will ask for a password to protect the key. Type a password.

- **cat data.aes.b64**
-

```
File Actions Edit View Help
root@kali: ~/multi
# mkdir multi
root@kali: ~/multi
# cd multi
root@kali: ~/multi
# echo "Hello World1">data.txt
root@kali: ~/multi
# cat data.txt
Hello World1
root@kali: ~/multi
# openssl enc -aes-256-cbc -base64 -in data.txt -out data.aes.b64
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
root@kali: ~/multi
# cat data.aes.b64
U2FsdGVkX19cHzZs0qckdeAmBS9z41n+7r7AIzKJYVU=
root@kali: ~/multi
#
```

First decode the encoded file and then decrypt the encrypted file using openssl and check the content:

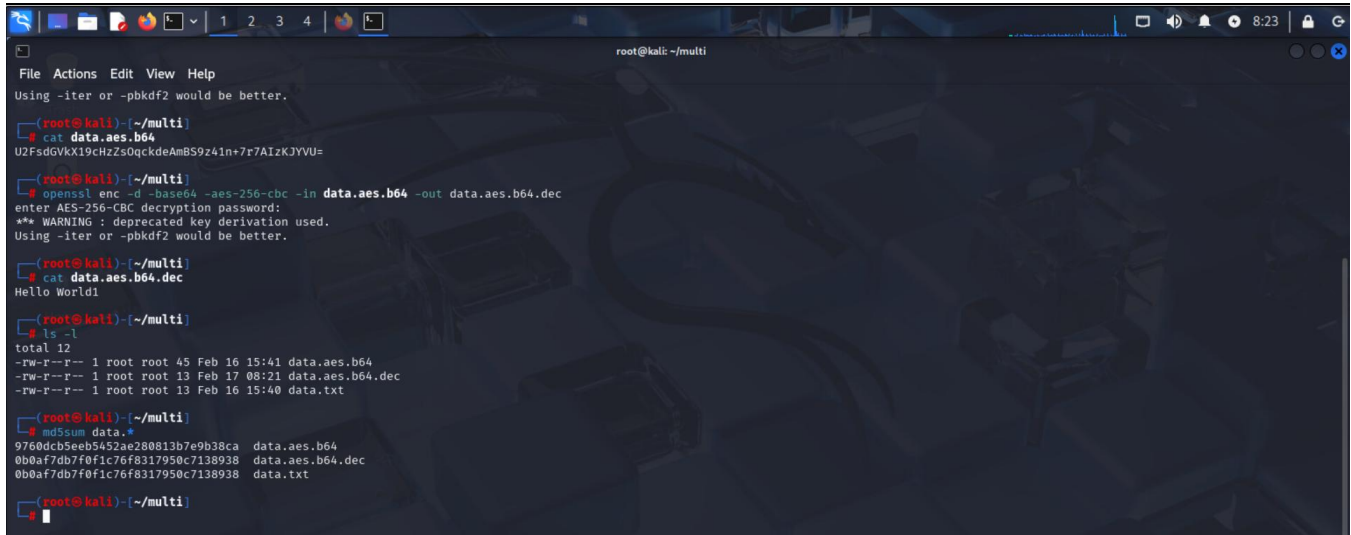
- `openssl enc -d -base64 -aes-256-cbc -in data.aes.b64 -out data.aes.b64.dec`
 - `cat data.aes.b64.dec`
-

```
File Actions Edit View Help
root@kali: ~/multi
# echo "Hello World1">data.txt
root@kali: ~/multi
# cat data.txt
Hello World1
root@kali: ~/multi
# openssl enc -aes-256-cbc -base64 -in data.txt -out data.aes.b64
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
root@kali: ~/multi
# cat data.aes.b64
U2FsdGVkX19cHzZs0qckdeAmBS9z41n+7r7AIzKJYVU=
root@kali: ~/multi
# openssl enc -d -base64 -aes-256-cbc -in data.aes.b64 -out data.aes.b64.dec
enter AES-256-CBC decryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
root@kali: ~/multi
# cat data.aes.b64.dec
Hello World1
root@kali: ~/multi
#
```

List the files and compare the md5 hashes:

- `ls -l`

- `md5sum data.*`



```
root@kali: ~/multi
File Actions Edit View Help
Using -iter or -pbkdf2 would be better.

(root@kali)~/multi
$ cat data.aes.b64
UZFsdGVkX19chZs0qckdeAmBS9z4ln+7r7AIzKJYVU=

(root@kali)~/multi
$ openssl enc -d -base64 -aes-256-cbc -in data.aes.b64 -out data.aes.b64.dec
enter AES-256-CBC decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(root@kali)~/multi
$ cat data.aes.b64.dec
Hello World!

(root@kali)~/multi
$ ls -l
total 12
-rw-r--r-- 1 root root 45 Feb 16 15:41 data.aes.b64
-rw-r--r-- 1 root root 13 Feb 17 08:21 data.aes.b64.dec
-rw-r--r-- 1 root root 13 Feb 16 15:40 data.txt

(root@kali)~/multi
$ md5sum data.*
9760dcb5eeb5452ae280813b7e9b38ca data.aes.b64
0b0af7db7f0f1c76f8317950c7138938 data.aes.b64.dec
0b0af7db7f0f1c76f8317950c7138938 data.txt

(root@kali)~/multi
$
```

Module Assessment

Question 4 - What encoding technique was used on encryption? Compare the screenshot of encrypted file from aes encryption (part two – data.aes) and encrypted file from multiple operation (part three – data.aes.b64)? What is the difference between two files.

Explanation of Encoding Technique

The technique used was Base64 encoding after AES encryption.

- AES-256-CBC: Symmetric encryption using a 256-bit key with Cipher Block Chaining.
- Base64 Encoding: Converts binary data into ASCII characters (readable text) for easier transport and storage.

Difference between data.aes (AES only) and data.aes.b64 (AES + Base64):

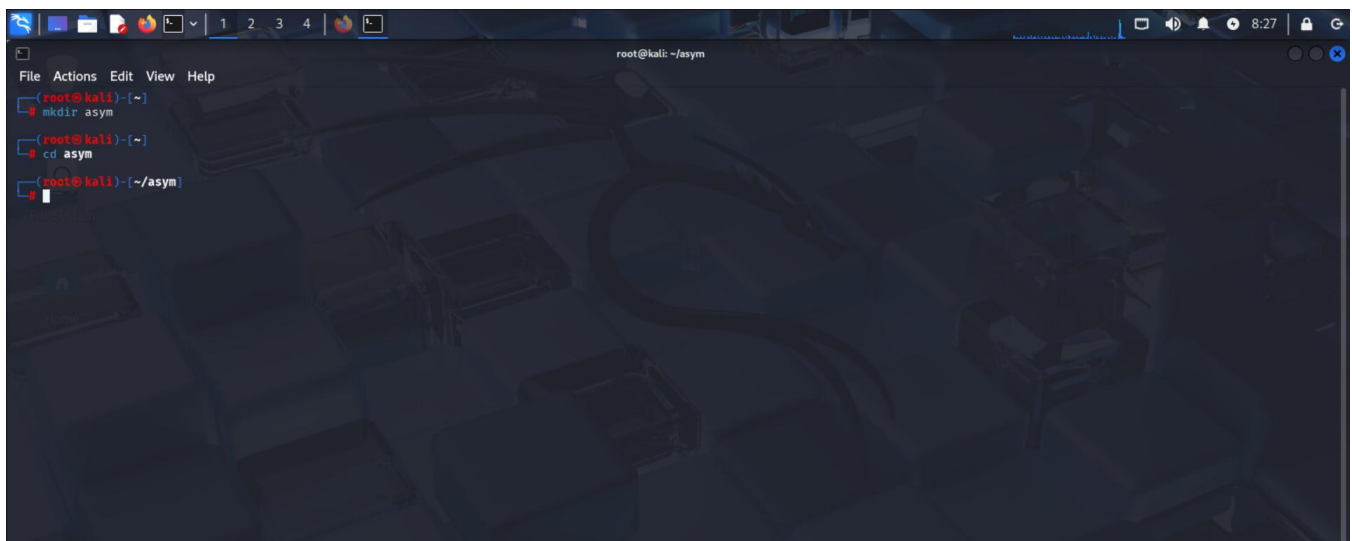
- data.aes: Binary file (unreadable) with raw encrypted content.
- data.aes.b64: Text file with Base64-encoded encrypted content, making it readable (though not understandable without decryption).

Module Activity Description:

Part Four: Asymmetric Encryption

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- `mkdir asym`
 - `cd asym`
-



Create a file and check the content:

- `echo "Hello World! > data.txt"`
 - `cat data.txt`
-

```
root@kali: ~/asym
File Actions Edit View Help
(root@kali)~# mkdir asym
mkdir: cannot create directory 'asym': File exists
(root@kali)~# cd asym
(root@kali)~/asym# echo "Hello World1">data.txt
(root@kali)~/asym# cat data.txt
Hello World1
(root@kali)~/asym#
```

Generate a private key using OpenSSL and check the content of the key:

- `openssl genpkey -algorithm rsa -pkeyopt rsa_keygen_bits:1024 -out privatekey.pem`
 - `cat privatekey.pem`
-

```
(root@kali)~/asym# openssl genpkey -algorithm rsa -pkeyopt rsa_keygen_bits:1024 -out privatekey.pem
.....
.....
(root@kali)~/asym# cat privatekey.pem
-----BEGIN PRIVATE KEY-----
MIICdgIBADANBgkqhkiG9w0BAQFAASCAmAwggJcAgEAAoGBAMZ/NVCJzofi/XS8
MCIKCIRL/+D26c2n+oR18z8Wp7VQ0R1kXfS9fXmPVEmpbtD5z9c1nDLDYfNKYY
Xee60rwcF+4AC7VaiCvSyHfx2k5+q0FUq6mrsvWRyUG7gmsQe/w7vNh4k+c4M3z
Ph7Mcs7LbHSFwKsL+bbB2bc1E8mBAGMBAECgYEAw1gNL+Vuy2wn71JrHB5JrQ7e
Enp1DXJqEkybqRw5HXZwFoKao7dEy+Flwh1hS0Lxm29XYyyt1ELmp1bEa4AV
yJMXDt0hhoU3aYNqWm1Ge71A2ofnL/kwA1zAEm//TXS/3mTx3kYtc9SegyLMPAU
Rp6n9NY12YcdK0oFL8ECQQDmlURDEAFcEckBczDlH0e01sD5pt7FnP7H6rXUrtb3v
fvj0KEQZewHYmmERcr4QwVANE5YF7P1PiY14UXKDzUPAKEA3KHFw9AYJSwe5MOE
bcpnGEZeagob40Jh9357UHmdeTw7evx0CY+0y1gTFGzGAHYa4xqv//IoK5KHB
uhe4bwJASNgq7VOP3pa/f65vzVuQyJoBaXLvLQu7h++h2y4JvM3z/4UcxLF2TAPH
37Tghbbs2joxasLUZ3SZ/LEBub7FkQJAFuV00Zg5uTnP5uuHRAf4e2NqDtUHud16
KrhVZFKycR6C/g+E1UeV2FwKz2wCVFb61+oUy2sPffRU73FLdu1TwJAPfPKT1Mq
561xwMI3ySVf6M5vPTLj0jrE8Ne+B4ghdBFVX+gM00oeOotZwGgyqE+nWqEqwXk
QmPpkJQHc3fkQ=
-----END PRIVATE KEY-----
(root@kali)~/asym#
```

Generate the public key associated with the private key and check the content of the key:

- `openssl rsa -pubout -in privatekey.pem -out publickey.pem`

- cat publickey.pem

```
File Actions Edit View Help
root@kali: ~/asym
-----BEGIN PUBLIC KEY-----
MIICdgIBADANBgkqhkiG9w0BAQEFAAQCAwAgGjCjAgEAAoGBAMZ/MVC3zofi/X58
MCIKCIRL/+DZ6c2n+oR18z8wp7VQ0R1kXf59fXmPVEmXpbtD5z9cTnDLLOYfNKYY
Xee60rwcF+4AC7ValcvSyHfX2k5+qDFUq6mrvsWRyUG7gmsQe/w7vNh4k+c4M3z
Ph7Mcs7LbHSFwKsL+bbB2bc1E8mBAGMBAACgYEAw1gML+Vuy2wn7IJrHBSjrQ7e
EnpiDXJyqEkrybqR8W5HXZWoXKa0TdEy+fwhh1hSolxm29XYyyvtELmp1bEa4AAV
yjMXDtDhhoU3aYnqVwniGe7IA2ofnL/kwA1zAEm//TX5/JmTx3kYtC95egyLMPAu
Rp6n9NY12YcdK0oFL8EQQmURDEAFECk8czDlH0e0iSDSpt7Fnp7HgrXurthb3v
fvj0KEQZEwhYmmERa4QwVAnSVF7PlD1Y14UXKDUPAKA3KHf9AYJ3we5MOE
bcpgGEZaogob40Jjh9357UHmxdeTw7evx0CY+Oy1gTfgcWGAHyaXqev//ioK5KHB
uhe4bwJASNggtVOP3pa/f65vzVuQy3oBaXLvLq7h++h2y4JvM3z/4UcxlF2TAPH
37Tghbbs2joxas1U2J5Z/LEBub7Fk0JAFuV00Zg5uTnF5uuHRAf4e2NqDtUHud16
KrhVZFKvcR6C/g+E1UeV2FwKr2wCvFb61+oUy2sPffRU73FLdu1TwJAPFKT1Mq
561xwhI3ysVf6M5vPTlJ0jrE8Ne+B4ghdBFXV+gM0OoeOotzwGgyE+nWQEqwXk
QmMpkJQHCA3fkQ=
-----END PUBLIC KEY-----

root@kali:~/asym# openssl rsa -pubout -in privatekey.pem -out publickey.pem
writing RSA key

root@kali:~/asym# cat publickey.pem
-----BEGIN PUBLIC KEY-----
MIIGFMAoCCSgQGA1UoQEAQAAAGNADCB1QKBgQDGFzVoi6H4v1+fDAicGIES//g
2enNp/qEdFM/Fqe1UNEedZF30vX15j1R3l6W70+c/XC3wyw2Hz5mGF3nutK8HH/U
AAu1WtXL70sh38dpOfajhVkuqp77FkclBu4JrEHV8072YeJPnODN8z4ezHLO5Wx0
hcCrC/m2wdm3NRPJgQIDAQAAB
-----END PUBLIC KEY-----

root@kali:~/asym#
```

Preview the private key:

- openssl rsa -text -in privatekey.pem

```
File Actions Edit View Help
root@kali: ~/asym
root@kali:~/asym# openssl rsa -text -in privatekey.pem
Private-Key: (1024 bit, 2 primes)
modulus:
00:c6:7f:35:50:89:ce:87:e2:fd:7e:7c:30:22:0a:
08:84:4b:ff:e0:d9:e9:cd:a7:fa:84:75:f3:3f:16:
a7:b5:50:d1:1d:64:5d:f4:bd:7d:79:8f:54:49:97:
a5:bb:43:e7:3f:5c:22:70:cb:2c:36:1f:34:a6:18:
5d:e7:ba:d2:bc:1c:7f:ee:00:0b:b5:5a:d5:cb:ef:
4b:21:df:c7:69:39:fa:a3:85:52:a6:a6:ae:fb:16:
47:25:06:ee:09:ac:41:ef:f0:ee:f3:61:e2:4f:9c:
e0:cd:f3:3e:1e:cc:72:ce:e5:6c:74:85:c0:ab:0b:
f9:b6:c1:d9:b7:35:13:c9:81
publicExponent: 65537 (0x10001)
privateExponent:
00:c3:58:0d:2f:e5:6e:cb:6c:27:ec:82:6b:1c:14:
a3:ad:0e:de:12:7a:62:00:72:72:a8:49:2b:c9:ba:
91:f1:6a:47:5d:95:98:a1:72:9a:39:37:44:cb:c7:
e1:c2:19:61:4a:89:71:9b:6f:57:63:2b:f2:b4:42:
e6:a7:56:c4:6b:80:15:ca:33:17:0e:d0:e1:86:85:
37:69:83:6a:57:09:e2:19:ee:e2:03:6a:1f:9c:bf:
e4:c0:08:b3:00:49:bf:fd:35:f9:fc:99:93:c7:79:
18:21:cf:79:7a:0c:8b:30:f0:2e:46:9e:a7:f4:d6:
22:d9:80:9d:28:ea:05:2f:c1
prime1:
00:a6:51:10:c4:01:f1:02:28:17:33:0e:51:f4:7b:
48:ac:0d:2a:6d:ec:59:cf:ec:71:ab:5d:4a:ed:85:
bd:ef:7e:f8:f4:28:44:19:13:01:d8:9a:61:11:72:
b6:b8:43:05:40:35:e4:98:17:b3:e5:3c:8c:88:e1:
45:ca:0f:35:0f
prime2:
00:dc:al:c5:c3:d0:18:25:2c:1e:e4:c3:84:6d:ca:
```

```
root@kali: ~/asym
File Actions Edit View Help
prime2:
00:dc:a1:c5:c3:d0:18:25:2c:1e:e4:c3:84:6d:ca:
67:18:46:5e:6a:0a:1b:e3:42:49:87:dd:f9:ed:41:
e6:c5:d7:93:c3:b7:af:c7:40:98:f8:ec:b5:81:37:
c6:cf:01:80:1f:26:b8:c6:ab:ff:fc:8a:0a:e6:41:
c1:ba:17:b8:6f
exponent1:
48:d8:2a:ed:53:8f:de:96:bf:7f:ae:6f:cd:5b:90:
c8:9a:01:69:72:ef:95:0b:bb:87:ef:a1:db:2e:09:
bc:cd:f3:ff:85:1c:c4:b1:76:4c:03:c7:df:b4:e0:
85:b6:ec:da:3a:31:6a:c9:54:64:94:99:fe:51:01:
b9:be:c5:91
exponent2:
16:65:4e:d1:98:39:b9:39:cf:e6:eb:87:45:a1:78:
7b:63:6a:0e:d5:07:b9:d8:ba:2a:b8:55:64:52:af:
71:1e:82:fe:0f:84:d5:47:95:d8:57:30:2a:bd:b0:
0a:f1:5b:eb:5f:a8:53:2d:ac:3d:f7:d1:53:bd:c5:
2d:db:a2:4f
coefficient:
3d:f3:ca:4e:23:2a:e7:ad:71:5a:12:37:c9:25:5f:
e8:ce:6f:3d:39:63:d2:3a:c4:f0:d7:be:07:88:21:
74:11:55:5f:e8:0c:d0:ea:1e:3a:8b:73:c0:68:32:
a8:4f:a7:5a:a1:2a:bb:05:e4:42:63:29:90:94:07:
70:0d:df:91
writing RSA key
-----BEGIN PRIVATE KEY-----
MIICDgIBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBAMZ/NVCJzofi/XS8
MCICIRL/+D26c2n+oR18z8Wp7VQ0R1kXf59fXmPVEmXpbtD5z9c1nDLDLYfNKYY
Xee60rwcF+4AC7Va1cVvSyHf2k5+qDFUq6mrvsWRyUG7gmsQe/w7vNh4k+c4M3z
Ph7Mcs7LbHSFwKsL+bbB2bc1E8mBAGMBAACgYEAWiGNL+Vuy2wn7IJrHBSJrQ7e
EnpIDXJyqEkrybqR8W5HXZWyOxKaOTdEy+fwhhLhSolxm29XYyvytELmp1bEa4AV
yJMXDtDhhoU3aYnQVwniGe71A2ofnL/kwA1zaEm//TX5/JmTx3kYlC9SegyLMPAU
Rpn9MY12CdK00F18ECQ0dUJRDE4FECK8ezDLH0e01s0Sp17FnPPHGxU0rLhb3v
fv30KEQZEwhYmmERcraQwVAnSVF7P1P1y14UUXKD2UPAKE3KHfW9AY3Swe5MOE
bcpnGEZaagob40Jh9357Uhmxdetw7evx0CY+Oy1gTfGzWGAHYa4xqv//IoK5KHB
uhe4bwJASNgq7VOP3pa/f65vzVuQyJoBaXLvLQu7h++h2y4JyM3Z/4UcxLF2TAPH
37Tghbbs2joxasLUZJSZ/LEBub7FkQJAFuV00Zg5uTnPsuuHRAf4e2NqDtUHudi6
KrhVZFKvcR6C/g+E1UeV2FcwKr2wCvFb61+oUy2sPffRU73FLduiTwJAPFKTIQm
561xwhI3ysVF6M5vPTLj0jrE8Ne+B4ghdBFXV+gM00oeOotzwGgyqE+nWqEqwXk
QmMpkJQHCa3fkQ=
-----END PRIVATE KEY-----
```

```
-----BEGIN PRIVATE KEY-----
MIICDgIBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBAMZ/NVCJzofi/XS8
MCICIRL/+D26c2n+oR18z8Wp7VQ0R1kXf59fXmPVEmXpbtD5z9c1nDLDLYfNKYY
Xee60rwcF+4AC7Va1cVvSyHf2k5+qDFUq6mrvsWRyUG7gmsQe/w7vNh4k+c4M3z
Ph7Mcs7LbHSFwKsL+bbB2bc1E8mBAGMBAACgYEAWiGNL+Vuy2wn7IJrHBSJrQ7e
EnpIDXJyqEkrybqR8W5HXZWyOxKaOTdEy+fwhhLhSolxm29XYyvytELmp1bEa4AV
yJMXDtDhhoU3aYnQVwniGe71A2ofnL/kwA1zaEm//TX5/JmTx3kYlC9SegyLMPAU
Rpn9MY12CdK00F18ECQ0dUJRDE4FECK8ezDLH0e01s0Sp17FnPPHGxU0rLhb3v
fv30KEQZEwhYmmERcraQwVAnSVF7P1P1y14UUXKD2UPAKE3KHfW9AY3Swe5MOE
bcpnGEZaagob40Jh9357Uhmxdetw7evx0CY+Oy1gTfGzWGAHYa4xqv//IoK5KHB
uhe4bwJASNgq7VOP3pa/f65vzVuQyJoBaXLvLQu7h++h2y4JyM3Z/4UcxLF2TAPH
37Tghbbs2joxasLUZJSZ/LEBub7FkQJAFuV00Zg5uTnPsuuHRAf4e2NqDtUHudi6
KrhVZFKvcR6C/g+E1UeV2FcwKr2wCvFb61+oUy2sPffRU73FLduiTwJAPFKTIQm
561xwhI3ysVF6M5vPTLj0jrE8Ne+B4ghdBFXV+gM00oeOotzwGgyqE+nWqEqwXk
QmMpkJQHCa3fkQ=
-----END PRIVATE KEY-----

root@kali) ~ - /asym
```

Preview the public key:

- openssl pkey -in publickey.pem -pubin -text

```
root@kali) ~ - /asym
File Actions Edit View Help
bcpnGEZaagob40Jh9357Uhmxdetw7evx0CY+Oy1gTfGzWGAHYa4xqv//IoK5KHB
uhe4bwJASNgq7VOP3pa/f65vzVuQyJoBaXLvLQu7h++h2y4JyM3Z/4UcxLF2TAPH
37Tghbbs2joxasLUZJSZ/LEBub7FkQJAFuV00Zg5uTnPsuuHRAf4e2NqDtUHudi6
KrhVZFKvcR6C/g+E1UeV2FcwKr2wCvFb61+oUy2sPffRU73FLduiTwJAPFKTIQm
561xwhI3ysVF6M5vPTLj0jrE8Ne+B4ghdBFXV+gM00oeOotzwGgyqE+nWqEqwXk
QmMpkJQHCa3fkQ=
-----END PRIVATE KEY-----

root@kali) ~ - /asym
$ openssl pkey -in publickey.pem -pubin -text
-----BEGIN PUBLIC KEY-----
MIGfMA0GCsGqSIb3DQEBAQUAA4GNADCBiQKBgQDGFzVQ1c6H4v1+FDAICg1ES//g
2enNp/qEdFM/Fqe1UNEdF30vX15j1R3Jl6W7Q+c/XCJwyw2HzSmGF3nutK8HH/u
AAu1WtXL70sh38dpQfjvhvKupq77FkLBU4JrEHv8072YeJPn00N8z4ezHLOSwx0
hcCrC/m2wdm3NRPJgQ1DAQAB
-----END PUBLIC KEY-----
Public-Key: (1024 bit)
Modulus:
00:c6:7f:35:50:89:ce:87:e2:fd:7e:7c:30:22:0a:
08:84:4b:ff:e0:d9:e9:cd:a7:fa:84:75:f3:3f:16:
a7:b5:50:d1:1d:64:5d:f4:bd:7d:79:8f:54:49:97:
a5:bb:43:e7:3f:3c:22:70:cb:2c:36:1f:3a:a6:18:
5d:e7:ba:db:bc:1c:7f:ee:00:0b:b5:5a:c5:cb:ef:
4b:21:df:c7:69:39:fa:33:85:52:ae:a6:ae:fb:16:
47:25:06:ee:09:ac:41:ef:f0:ee:f3:61:e2:4f:9c:
e0:cd:f3:3e:1e:cc:72:ce:5e:6c:74:85:c0:ab:0b:
f9:b6:c1:d9:b7:35:13:c9:81
Exponent: 65537 (0x10001)

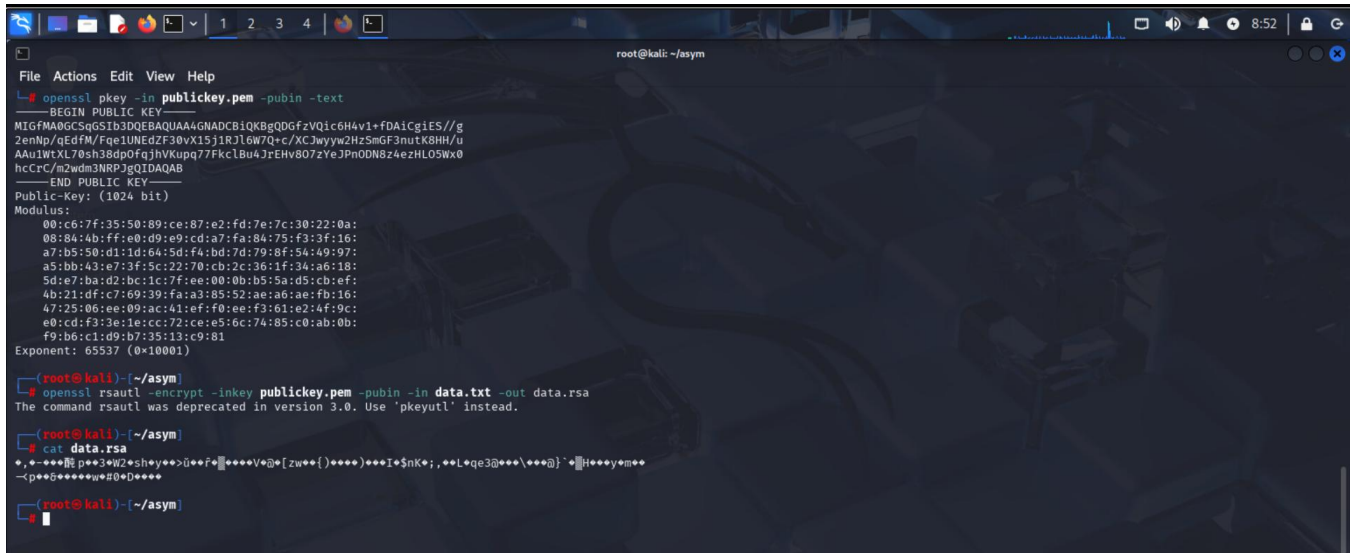
root@kali) ~ - /asym
```

Encrypt the file you created and check the content:

- **`openssl rsautl -encrypt -inkey publickey.pem -pubin -in data.txt -out data.rsa`**
- **`cat data.rsa`**

- **openssl rsautl -encrypt -inkey publickey.pem -pubin -in data.txt -out data.rsa**

- **cat data.rsa**



Decrypt the encrypted file and check the content:

- **`openssl rsautl -decrypt -inkey privatekey.pem -in data.rsa -out data.rsa.dec`**
- **`cat data.rsa.dec`**

- **openssl rsautl -decrypt -inkey privatekey.pem -in data.rsa -out data.rsa.dec**

- **cat data.rsa.dec**

```
root@kali: ~/asym
File Actions Edit View Help
Modulus:
00:c6:7f:35:50:89:ce:87:e2:fd:7e:7c:30:22:0a:
08:84:4b:ff:e0:d9:e9:cd:a7:fa:84:75:f3:3f:16:
a7:b5:50:d1:1d:64:5d:f4:bd:7d:79:8f:54:49:97:
a5:bb:43:e7:3f:5c:22:70:cb:2c:36:1f:34:a6:18:
5d:e7:ba:d2:bc:1c:7f:ee:00:0b:b5:5a:d5:cb:ef:
4b:21:df:c7:69:39:fa:a3:05:52:ae:a6:ae:fb:16:
47:25:06:ee:09:ac:41:ef:f0:ee:f3:61:e2:4f:9c:
e0:cd:f3:3e:1e:cc:72:ce:e5:6c:74:85:c0:ab:0b:
f9:b6:c1:d9:b7:35:13:c9:81
Exponent: 65537 (0x10001)

root@kali) ~/asym
# openssl rsautl -encrypt -inkey publickey.pem -pubin -in data.txt -out data.rsa
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.

root@kali) ~/asym
# cat data.rsa
+-----+
pe3W2shyey++u++f+Ved@{zw++{)+++++Ie$Nk+;,,eLeqe3@++\+++@}~*H++yem++
~p++5++++w@#D++++

root@kali) ~/asym
# openssl rsautl -decrypt -inkey privatekey.pem -in data.rsa -out data.rsa.dec
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.

root@kali) ~/asym
# cat data.rsa.dec
Hello World!

root@kali) ~/asym
#
```

List the files and compare the md5 hashes:

- `ls -l`

- `md5sum data.*`

```
root@kali) ~/asym
# ls -l
total 20
-rw-r--r-- 1 root root 128 Feb 17 08:51 data.rsa
-rw-r--r-- 1 root root 13 Feb 17 08:57 data.rsa.dec
-rw-r--r-- 1 root root 13 Feb 17 08:30 data.txt
-rw-r--r-- 1 root root 916 Feb 17 08:39 privatekey.pem
-rw-r--r-- 1 root root 272 Feb 17 08:44 publickey.pem

root@kali) ~/asym
# md5sum data.*
c88997d0077749760cb9e907963271ab data.rsa
0b0af7db7f0f1c76f8317950c7138938 data.rsa.dec
0b0af7db7f0f1c76f8317950c7138938 data.txt

root@kali) ~/asym
#
```

Module Assessment

Question 5 - What is a private key and what is a public key? What are the name of files used to encrypt and decrypt data.txt in this section?

1. What is a Private Key?

- A **private key** is a confidential component of asymmetric encryption.
- It is used for **decryption** and **signing** messages.

- It must be kept secure and never shared publicly.

2. What is a Public Key?

- A **public key** is derived from the private key.
- It is used for **encryption** and **verification**.
- It can be shared openly without security risks.

Files Involved in Encryption/Decryption

Operation	File Name	Purpose
Input File	data.txt	Original plaintext file
Private Key	privatekey.pem	For decrypting data
Public Key	publickey.pem	For encrypting data
Encrypted File	data.rsa	RSA-encrypted version of data
Decrypted File	data.rsa.dec	Decrypted plaintext file

Module Activity Description:

Part Five: Digital Signatures

1. Paste the screenshot of different hash values of the data file (data.txt).

- `openssl dgst -md5 data.txt`

- `openssl dgst -sha1 data.txt`

- `openssl dgst -sha256 data.txt`

```
root@kali: ~/asym
File Actions Edit View Help
rm primary.pem

(root@kali)~/asym
ls -l
total 20
-rw-r--r-- 1 root root 128 Feb 17 08:51 data.rsa
-rw-r--r-- 1 root root 13 Feb 17 08:57 data.rsa.dec
-rw-r--r-- 1 root root 13 Feb 17 08:30 data.txt
-rw-r--r-- 1 root root 916 Feb 17 08:39 privatekey.pem
-rw-r--r-- 1 root root 272 Feb 17 08:44 publickey.pem

(root@kali)~/asym
md5sum data.*
c88997d0077749760cb9e907963271ab data.rsa
0b0af7db7f0f1c76f8317950c7138938 data.rsa.dec
0b0af7db7f0f1c76f8317950c7138938 data.txt

(root@kali)~/asym
openssl dgst -md5 data.txt
MD5(data.txt)= 0b0af7db7f0f1c76f8317950c7138938

(root@kali)~/asym
openssl dgst -sha1 data.txt
SHA1(data.txt)= 0aa4f38d72dc55ff0617bdce4796c9f9022a6bef

(root@kali)~/asym
openssl dgst -sha256 data.txt
SHA2-256(data.txt)= 8c88baf48813956d77c70b2c3ba29f347a3ae8b454658df31355ff8be3bc3576

(root@kali)~/asym
```

Sign the file and check the content of the signature:

- `openssl dgst -sha256 -sign privatekey.pem -out signature.bin data.txt`
- `ls -l`

```
root@kali: ~/asym
File Actions Edit View Help
c88997d0077749760cb9e907963271ab data.rsa
0b0af7db7f0f1c76f8317950c7138938 data.rsa.dec
0b0af7db7f0f1c76f8317950c7138938 data.txt

(root@kali)~/asym
openssl dgst -md5 data.txt
MD5(data.txt)= 0b0af7db7f0f1c76f8317950c7138938

(root@kali)~/asym
openssl dgst -sha1 data.txt
SHA1(data.txt)= 0aa4f38d72dc55ff0617bdce4796c9f9022a6bef

(root@kali)~/asym
openssl dgst -sha256 data.txt
SHA2-256(data.txt)= 8c88baf48813956d77c70b2c3ba29f347a3ae8b454658df31355ff8be3bc3576

(root@kali)~/asym
openssl dgst -sha256 -sign privatekey.pem -out signature.bin data.txt

(root@kali)~/asym
ls -l
total 24
-rw-r--r-- 1 root root 128 Feb 17 08:51 data.rsa
-rw-r--r-- 1 root root 13 Feb 17 08:57 data.rsa.dec
-rw-r--r-- 1 root root 13 Feb 17 08:30 data.txt
-rw-r--r-- 1 root root 916 Feb 17 08:39 privatekey.pem
-rw-r--r-- 1 root root 272 Feb 17 08:44 publickey.pem
-rw-r--r-- 1 root root 128 Feb 17 09:44 signature.bin

(root@kali)~/asym
```

Receiver to verify the signature:

- `openssl dgst -sha256 -verify publickey.pem -signature signature.bin data.txt`

- `ls -l`

```
root@kali: ~/asym
SHA2-256(data.txt)= 8c88baf48813956d77c70b2c3ba29f347a3ae8b454658df31355ff8be3bc3576

root@kali:~/asym# openssl dgst -sha256 -sign privatekey.pem -out signature.bin data.txt

root@kali:~/asym# ls -l
total 24
-rw-r--r-- 1 root root 128 Feb 17 08:51 data.rsa
-rw-r--r-- 1 root root 13 Feb 17 08:57 data.rsa.dec
-rw-r--r-- 1 root root 13 Feb 17 08:30 data.txt
-rw-r--r-- 1 root root 916 Feb 17 08:39 privatekey.pem
-rw-r--r-- 1 root root 272 Feb 17 08:44 publickey.pem
-rw-r--r-- 1 root root 128 Feb 17 09:44 signature.bin

root@kali:~/asym# openssl dgst -sha256 -verify publickey.pem -signature signature.bin data.txt
Verified OK

root@kali:~/asym# ls -l
total 24
-rw-r--r-- 1 root root 128 Feb 17 08:51 data.rsa
-rw-r--r-- 1 root root 13 Feb 17 08:57 data.rsa.dec
-rw-r--r-- 1 root root 13 Feb 17 08:30 data.txt
-rw-r--r-- 1 root root 916 Feb 17 08:39 privatekey.pem
-rw-r--r-- 1 root root 272 Feb 17 08:44 publickey.pem
-rw-r--r-- 1 root root 128 Feb 17 09:44 signature.bin
```

Module Assessment

Question 6 - What was used to verify the signature of data.txt. Will verification fail if any of the file – signature.bin, publickey.pem, data.txt changes?

What was used to verify the signature?

- The public key (publickey.pem) was used to verify the signature.
- OpenSSL calculates the SHA256 hash of data.txt, decrypts signature.bin with the public key, and compares the two hashes.
- If the hashes match, verification succeeds.

Will verification fail if any file changes?

Yes, verification will fail if any of the following changes:

1. data.txt: Changes in content will alter the hash and cause verification failure.
2. signature.bin: If tampered with, it will not correctly match the file's hash.
3. publickey.pem: If replaced with a different key, the signature will not decrypt correctly.